Deposits at War:

How Outside Deposit Competition During a Crisis Affects Banks

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Abstract

What are the short- and long-run effects on banks of deposit competition outside of the banking sector? Using a unique historical setting with an exogenous shock to local deposit markets, this paper presents new causal evidence on commercial banks' deposit pricing and funding structure. Depositors in the Netherlands could historically open an account at the Dutch central bank. These accounts sat idle until the outbreak of World War I, which temporarily drove deposit balances at the central bank to unprecedented levels. More exposed banks, with a larger share of branches in central bank cities, raised their short-term deposit rates further and were more likely to invert their deposit term structure. Deposits returned to commercial banks once the crisis subsided, but exposed banks relied less on deposits in their funding mix and expanded their lending less in the following years. The findings suggest that short-lived deposit competition during a crisis can reshape how banks fund themselves.

Keywords: Banking, financial stability, financial history, deposit competition

JEL Codes: E43, E58, G21

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Despite being runnable, retail deposits are commercial banks' cheapest and most stable source of funding. The costs and stability of banks' funding are key for their ability to extend credit to the real economy. The rise of outside deposit competition, i.e. depositing alternatives originating outside of the banking sector such as stablecoins and central bank digital currencies, can threaten banks' deposit franchise. Theory predicts that competition for bank deposits can lead to instability and disintermediation (Andolfatto, 2021, Keister and Sanches, 2023, Whited et al., 2023). Empirically, changes in deposit competition are rarely exogenous to bank or local economic conditions, making causal identification difficult. Moreover, empirical evidence of deposit competition during crisis times, when it matters most, is largely absent. This paper studies an exogenous shock to deposit markets and provides empirical evidence on the effects of outside competition on commercial banks' deposit pricing and funding decisions.

It is unclear ex ante how outside deposit competition will affect banks. Retail depositors are known to be sticky and slow to switch (Drechsler et al., 2017, Fleckenstein and Longstaff, 2024, Yankov, 2024). Absent any remuneration or additional services provided by outside options, commercial banks may remain the more attractive option. In normal times, depositors likely prioritize returns and convenience over safety. During crisis times, however, safety concerns may dominate. This can trigger a flight to safety toward assets initially designed as a means of payment, which then unintentionally become a store of value.

Leveraging a unique and hand-collected dataset of daily advertisements in historical newspapers by 77 Dutch banks, this paper shows how banks responded differently to the outbreak of the First World War depending on their share of branches located in cities with a central bank office. The Dutch central bank historically accepted retail deposits, without remuneration or holding limits. Deposit markets were local, and otherwise similar banks differed in their exposure to central bank competition. More exposed banks increased their demandable deposit rate by an additional 46 basis points in the first weeks after the shock and were four times more likely to invert their deposit term structure. The price response disappeared once the crisis subsided. Despite deposits returning to commercial banks within months, more exposed banks relied less on deposits in their funding mix in the years after.

The Dutch central bank accepted deposits on current accounts in an attempt to nationalize and modernize the Dutch payment system from cash to a credit transfer system (*giro*). Due to a depositing fee and no remuneration, the accounts were mainly used by banks for making payments. The outbreak of the First World War in 1914 changed this: the imminent war threat triggered a short-lived bank run on deposits at commercial

banks and a deposit flight to the central bank.[1] DNB received an inflow of 30 million guilders ($f.$) of deposits, which is equivalent to 10% of the commercial banks' demandable deposits pre-war (Van Riel, 2021).[2]

The paper uses a difference-in-differences design with continuous treatment based on the overlap of a commercial bank's branches with the central bank offices. Banks with more branches located in a central bank city, had to raise their short-term deposit rates more following the outbreak of WWI. Compared to a non-exposed bank, a bank with all branches overlapping with central bank offices had to increase its deposit rate by an additional 35-55 basis points in the weeks after the shock. Compared to a pre-war average demandable rate of 2.88%, this is a large effect. More exposed banks were also four times as likely to adopt an inverted term structure of deposit pricing, meaning that they offered higher rates for shorter term deposits compared to longer maturity deposits. The results cannot be explained by bank characteristics or location differences pre-shock: the effect remains when controlling for bank size, city population, local unemployment and local bank competition. More exposed banks relied less on deposit funding in the years after the shock and consequently grew slower compared to low-exposed banks. Greater exposure to the central bank is correlated with slower lending growth, driven by a decrease in illiquid loans, consistent with the perception that deposit funding became unstable. Banks that had to borrow from the central bank had to do so at a higher rate.

The results show that central bank retail deposits, despite being designed as a means of payment, can unintentionally become a store of value in times of heightened uncertainty. This can reshape how commercial banks fund themselves. The results should be interpreted as a reassessment by banks on the cost of deposit funding. Before the outbreak of the war, commercial banks viewed deposit funding as both cheap and stable. During the months of heightened uncertainty, the presence of a central bank office provided depositors with an extra outside option, amplifying withdrawal pressures. This made the bank consequently reassess the riskiness of deposit funding: the perceived cost of deposit funding increased. Deposit rate differentials disappeared and deposits flew back to all commercial banks once the panic subsided when war neutrality was assured. This reassessment still had long-lasting effects. Once the panic had faded, other types of bank funding became available again, while simultaneously deposit rates for high-exposed banks decreased. In the years after, banks more exposed to the central bank reduced their

[1] While the Netherlands remained neutral during WWI, the outbreak on the 28th of July 1914, the Amsterdam stock market closure on 29th of July and the Dutch mobilization on the 31st of July led to a widespread panic across the country. During the course of 1914 the threat of war diminished and the Dutch economy benefited substantially from the neutral status during the war (Petram, 2016; 't Hart et al., 1997).

[2] Notably, deposit outflows at commercial banks are a multiple of the inflows at the central bank, since many depositors choose cash over central bank accounts. The central bank presence thus serves as an additional outside option that can trigger withdrawals.

reliance on deposits. This suggests that those banks themselves chose to attract fewer deposits once alternative, but more expensive means of funding became available again. Exposure to the central bank was correlated with slower lending growth during the war.

The results provide one of the first empirical insights into the potential effects of deposits' competition for commercial banks. The exogenous shock to local deposit market competition isolates the causal effect of banks' market power on deposit pricing, bank funding and lending. While most literature has focused on the retail adoption of alternatives to bank deposits, such as stablecoins and CBDC, this paper provides a new perspective: commercial banks might also reassess their funding structure, irrespective of actual (persistent) outflows. The historical, non-digital, setting offers several advantages compared to digital payments setting today. In 1914, DNB accounts were physical and consequently slower and costlier to access than current digital safe assets. This paper leverages the physical distance to bank branches to identify exposed banks, which is more difficult to identify nowadays when all depositors can move their money around digitally. My findings likely represent a lower bound on the competitive effects of a central bank deposit alternative during financial stress. Moreover, the Netherlands did not have deposit insurance. This might alleviate concerns about behavioral distortions.[3] Since depositors of commercial banks and the central bank in 1914 were mostly wealthy households and firms, they are most comparable to uninsured depositors nowadays, which still make up around 55% (75%) of average European (US) commercial bank's balance sheet.[4] Moreover, even in the presence of deposit insurance, central bank accounts can offer additional convenience. The credibility of deposit insurance (Bonfim and Santos, 2023) and delays in settlement following bank failures might still lead to depositors preferring to move (some of) their deposits to the central bank, especially in crisis times.

Related Literature. This paper adds to two strands of literature. First, the paper contributes to the fast-growing literature on banks' market power in deposit markets. Kashyap, Rajan and Stein (2002) provide the foundation for banks' provision of liquidity while taking deposits. Drechsler, Savov and Schnabl (2017) build on this and show that local deposit markets with stronger competition have higher pass-through of monetary policy, that this deposit franchise allows banks with market power to charge lower rates, and that banks can hedge their interest rate risk by maturity transformation (Drechsler et al., 2021). Similar to this paper, Ahn and Brei (2024) show that deposit competition especially increases during crisis times. Apart from increasing rates, banks can attract and retain depositors in several ways: by branch expansion (Begenau and Stafford, 2025),

[3]See, for instance, Calomiris and Jaremski (2019)

[4]EBA Risk Assessment Report, June 2025:
https://www.eba.europa.eu/publications-and-media/publications/liabilities-funding-and-liquidity-1;
FED Statistical Release Asset and Liabilities of Commercial Banks in the United States, 13-02-2026:
https://www.federalreserve.gov/releases/h8/current/h8.pdf, respectively.

differentiating rates within the bank based on the market power in the local market (Drechsler et al., 2017), or differentiating between different types of depositors (Fleckenstein and Longstaff, 2024; Basten et al., 2026). In this paper, rate-setting is uniform and deposit markets are local by design. Rather than assuming an endogenous cross-sectional deposit market competition, the setting in this paper offers a clean, exogenous shock to local deposit-market competition. The variation in geographical exposure to central bank offices allows for testing the market-power mechanism through variation in competitive structure rather than inferred concentration or pricing behavior.

Drechsler, Savov, Schnabl and Wang (2026) show that the deposit franchise is in itself a source of fragility: it depends on depositors staying in the bank, which can trigger a run even when the bank's assets are fully liquid. This paper documents a repricing of the deposit franchise, but in a setting where the franchise was revised by banks rather than destroyed by depositors. Deposits returned within months, yet exposed banks did not raise their deposits compared to pre-war in the year after, which attracted fewer deposits and shrank their franchise. The contribution of this paper is therefore not that deposit competition raises rates, but that short-lived competition from a central bank that neither paid interest nor sought deposits persistently alters how banks price their deposit franchise.

Second, this paper adds to the literature on the flight-to-safety. Caballero and Krishnamurthy (2008) are among the first to document the phenomenon theoretically. While recent empirical work examines the flight-to-safety toward large banks (Cipriani et al., 2024) money market funds with central bank access (Gurrieri et al., 2025), bank deposits in general during geopolitical shocks (Anastasiou et al. 2025) and banks with the most credible deposit insurance (Bonfim and Santos 2023), this paper provides empirical evidence on flight directly to central bank deposits.

Theoretical research has documented the flight-to-safety to a CBDC.[5] Williamson (2022) shows that banking panics are more likely to occur under CBDC due to the strict dominance of CBDC over physical currency.[6] This paper shows that during times of heightened uncertainty, central bank retail deposits can provide an additional option for depositors to store their money, which especially affects the commercial banks most geographically exposed to the alternative. Similarly, Ahnert et al. (2025) show that banks will offer more attractive deposit contracts to retain depositor funding, which aligns with my find-

[5]For an overview of the theoretical and empirical literature on CBDC, see Ahnert et al. (2022).

[6]Similar to my empirical setting, Williamson (2022) assumes that CBDC is used by some depositors in the good state, but by all in the bad state. However, his theoretical model does assume that CBDC is interest-bearing, while in this paper's setting, the central bank depositing accounts do not pay any interest. Both in theoretical and empirical settings, disentangling the role of central bank retail deposits as means of payment or store of value is not straightforward. I will not attempt to isolate one of the two functions of money here.

ing that banks more exposed to the central bank increase their deposit rates more. While existing research has, in the absence of data, focused on the theoretical implications of CBDC on financial stability and potential disintermediation, this paper provides an empirical contribution of central bank competition for retail deposits. Keister and Monnet (2022) argue that a central bank might be better equipped to intervene timely during a bank run, while Brunnermeier and Niepelt (2019) point at the possibility for the central bank to fully pass through any deposits it may receive back to the commercial banking sector. I provide anecdotal evidence that commercial banks more exposed to the central bank tapped the discount window more often.

Finally, this paper adds to the financial history literature on central bank retail deposits. Antipa et al. (2026) provide an overview of central banks that accepted retail depositors in the past.[7] Related to this research are two other historical papers documenting depositor behavior during crisis when the central banks accepts retail deposits. Antipa and Rieder (2026) analyze the flight-to-safety to the Bank of England during the Overend & Gurney crisis in 1866, while Jorge-Sotelo (2026) shows that during the bank run following the political regime change in Spain in 1931, Banco d'España was a safe haven. My paper confirms the flight-to-safety of retail deposits to the central bank during crises, but adds to the current literature by causally estimating the effects of central bank competition for commercial banks' deposit pricing and funding structure. The Dutch central bank did not, contrary to the Bank of England and Banco d'España, actively compete for deposits. I can therefore study how commercial banks respond to the *mere existence* of a central bank alternative when an exogenous shock suddenly raises its attractiveness.

The rest of the paper is structured as follows. The first section the historical background. Section 2 covers the hand-collected dataset, while Section 3 explains the empirical approach. Section 4 presents the empirical results, Section 5 reconciles the findings in a conceptual framework and Section 6 concludes.

1 Historical Background

The Dutch central bank, *De Nederlandsche Bank* (DNB), was founded in 1814 in Amsterdam by King William I as a private institution with note issuance monopoly.[8] Until 1860, DNB was the largest financial institution in the Netherlands. Both financial and

[7] Grodecka-Messi & Zhang (2023, 2025) use the note-issuance monopoly of the central banks of Canada and Sweden, respectively, to illustrate the potential effects of central bank competition on the commercial banking sector.

[8] In nearly all other European countries in the nineteenth century, central banks coexisted with other note issuing banks. DNB was the first institution in The Netherlands to issue banknotes, though its charter did not assign it a legal monopoly, de facto the DNB was had the not issuance monopoly.

non-financial institutions could borrow from the central bank after official approval, either via discounting bills of exchange ("*wissels*") at the discount window, collateralized short-term borrowing with securities as collateral ("*beleeningen*"), or collateralized overdrafts on current account ("*voorschotten in rekening-courant*"). Most commercial banks ("*handelsbanken*") were founded after 1860, yet their initial growth was slow and their credit provision overly cautious (De Jong, 1967a). Only in the early twentieth century multi-branch networks of commercial banks began to develop (Jonker, 1995). Appendix A.1 provides more details on the development of DNB and the commercial banks before 1914.

1.1 The central bank offices

Until 1864, DNB was located only in Amsterdam. The Bank Act of 1863 changed this: it obliged DNB to open at least one branch (office, "*agentschap*") in each of its provinces. The branches were all headed by an 'agent' who had to pay in some of his own wealth and was personally liable for damages at the branch (Mooij and Prast, 2003). The branches were not independent on the head office in Amsterdam: they needed approval before accepting firms or individuals to their discount window and underwent daily checks of their books.[9]

The Dutch government mandated DNB to extend their network and credit facilities beyond its capital, which made it a true national bank (Van Riel, 2021). As a result, between 1864 and 1867, 14 branches opened across the Netherlands, making DNB the first Dutch bank with a national branch network (Kymmell, 1992).[10] From 1889 onwards, the branches accepted retail deposits in current accounts. In 1914, DNB had 20 office that functioned as depositing locations, including the head office in Amsterdam. The last branch was opened a decade before, in 1903. No branches were closed in the meantime. Figure B.1 shows the 20 DNB office locations and their corresponding responsible regions ("*ressorts*").

Importantly, DNB did not open a branch in every large city.[11] The selection of locations for branches was only partly dependent on the economic circumstances at the time: some

[9]One exception was the Rotterdam branch ("*Bijbank*"), the largest office outside of Amsterdam, which was a more stand-alone DNB office and did not require Amsterdam approval for extending credit. The central bank was free to open other regional correspondent offices, ("*correspondentschappen*", which could not accept deposits) at its discretion.

[10]The head office had existed since 1814 and was located in Amsterdam. The branches were located in Alkmaar, Almelo, Arnhem, Dordrecht, Groningen, The Hague, 's Hertogenbosch, Leeuwarden, Maastricht, Meppel, Middelburg, Rotterdam, Utrecht and Zwolle. DNB opened a branch in Leiden and Nijmegen in 1897; in Tilburg in 1892; and in Deventer in 1903. Enschede was a sub-branch from 1889 until 1915.

[11]Haarlem, Eindhoven and Breda are notable examples of cities that did not have a DNB branch.

cities in which a DNB branch opened had no substantial trade or industry, though the population was wealthier than the national average.[12]. In 1889 Leiden was promoted from regional correspondent to sub-branch despite having no borrowers at the discount window (apart from the regional correspondent itself the year before (Fase and Mooij, 2000)).[13] The location of DNB branches was thus partly determined by historical administrative decisions dating back to 1864. Still, the cities with a DNB branch were on average much larger compared to cities without a DNB branch, and the largest 4 cities (Amsterdam, Rotterdam, The Hague and Utrecht) do not have a counterfactual: there are no cities of comparable size without a central bank office. Using common support and matching procedures, the papers shows that results hold when controlling for city size and other economic characteristics.

1.2 Depositing at the central bank

DNB was initially not allowed to accept deposits or money from any other individual or institution than the state. One of the reason for refraining from (retail) deposits was to avoid competition with the "*cashiers*" in Amsterdam, which were bank-like institutions that primarily acted as safe-keepers of merchants, but also engaged in exchanging money, doing payments and extending credit (Kymmell, 1992). This changed with the renewal of its charter in 1838, when DNB allowed depositing in a current account, albeit at a fee of 1/2 ‰. Current accounts could be opened by anyone introduced to the bank. The accounts received no remuneration. Despite the low depositing fee (below that of the *cashiers*), very few non-financial firms and individuals deposited their money at DNB: in 1864, the total current accounts balances of non-financial firms and individuals was a meager $f.$ 415,000, or 0.39% of notes in circulation (De Jong, 1930).

From 1889 onward, the number of current account holders at DNB simultaneously increased with commercial banks depositors, albeit at a much slower rate. Some of the reasons for the reluctance to open a current account at DNB were 1) the depositing fee and 2) no remuneration, while commercial banks offered between 1-4%. Not surprisingly, the average total balance of all current account holders at DNB remained very modest: before the outbreak of the war it fluctuated around $f.$ 6 million, or 1% of its balance sheet size. Commercial bank deposits, on the other hand, consisted of around $f.$ 200 million in term deposits, and another $f.$ 300 million in savings and demandable deposits, which made up about 40 % of their total assets (Van Riel, 2021). Thus, the current accounts

[12] Arnhem and The Hague are examples of having little economic activity but wealthy inhabitants

[13] While the regional correspondent, a cashier, might have been to blame for the absence of discounting, but the year after Leiden becoming a subbranch the results were still disappointing in the view of its supervising agent (Fase and Mooij, 2000).

at DNB were not widely used beyond the financial sector before the outbreak of the war (De Jong, 1967b).

Similar to the current trade-off policy makers are facing nowadays, the Dutch Central Bank wanted to ensure the smooth functioning of the payment system, but prevent the outflow of deposits from commercial banks to the central bank. The president of DNB even wrote in the Economic Journal:

"*A fresh phenomenon has therefore appeared during the war; less and less credit is in demand from the bank of issue; its discounts and loans are reduced to unprecedented figures;* ***the public commences, on the contrary, to give the Bank a credit it did not ask for****, but which it must be open to receive as the central institution enjoying the confidence of all and sundry. The Bank becomes the great keeper of the reserves which the public is forced to lay aside for the time being, under the obligation, however, to place it at the disposal of the lenders at a moment's notice.*"[14]

The Dutch setting in 1914 thus allows to study how commercial banks respond to the *mere existence* of a central bank alternative when an exogenous shock suddenly raises its attractiveness. The historical set-up of retail deposits at the Dutch central bank are thus similar to current CBDC proposals that aim to facilitate payments, without actively competing for deposits with commercial banks.

1.3 The outbreak of World War I

On 28 July 1914, Austria-Hungary declared war on Serbia. The outbreak of the First World War triggered an unprecedented global financial panic, and the Netherlands was no exception.[15] The Amsterdam Stock Exchange closed on 29 July, soon followed by emergency government measures: on 31 July the gold standard was suspended and military mobilization was announced.

DNB implemented crisis interventions too. To ensure the continuation of the payment system, it removed the depositing fee for its current accounts on July 31.[16] The Bank saw substantial inflows of money on its current account from non-banks from the outbreak of the war. Figure 1 (a) shows the total daily current account balance at all DNB offices around the end of July as a percentage of its pre-war balance sheet size. Once the war breaks out, the current account balances start increasing rapidly with an inflow of $f.$ 30 million (or 10% of demandable commercial bank deposits (Van Riel, 2021). Despite the

[14] Vissering, 1917, p.168.

[15] See for descriptions of the New York and London Stock Exchange closures Silber (2007) and Roberts (2013), respectively.

[16] The depositing fee previously stood at 1/2 ‰. See Appendix A.4 for other war measures

simultaneous increase of banknotes in circulation, the share of current accounts to the total size of DNB's balance sheet increases from around 1% to maximum 10% on August 27 (see Figure 2 for the composition of the liability side of the balance sheet).

Mostly firms and households opened new accounts. Figure 1 (b) breaks down the number of current account holders in financial institutions and non-financial account holders (See Appendix B.2 for a breakdown by branch). Note that account holders can be both borrowers and lenders. While the financial institutions still dominated the current accounts of DNB before the war, at the end of 1914 the number of households and firms holding an account at DNB had increased fivefold to more than 1,500. The number of financial institutions also doubled from 379 to 768, but they mostly borrowed from the current account rather than deposited money (De Jong, 1967b).

1.4 Commercial bank runs in 1914

The threat that the neutrality of the Netherlands would not be honored and the consequent mobilization led to a widespread panic across the Netherlands. In this chaos, many depositors withdrew their money from the commercial banks (De Nederlandsche Bank, 1915).[17] The inflow at DNB was $f.$ 30 million in the first weeks, which is equivalent to 15% of demandable deposits at commercial banks pre-war (Van Riel, 2021). Many depositors also fled to cash: the total withdrawals at commercial banks exceeded the inflow at the central bank.[18] While the central bank geographical presence was thus not a prerequisite for a commercial bank run, it did lead to a stronger deposit rate response by banks facing this competition. The availability of central bank accounts thus offered an additional safe outside option for depositors.

The phenomena of receiving retail deposits at DNB branches was widespread and documented in the yearly reports of every branch.[19] The DNB president in his yearly report writes that the withdrawals at commercial banks were 'ungrounded' (De Nederlandsche

[17] I am grateful to Philip Fliers for sharing the DNB yearly reports from 1914-1916.

[18] Four of the five largest commercial banks of the country together lost more than $f.$ 90 million in deposits during the first days of the crisis, a quarter of their balance sheet size of 1913. Excerpts from the yearly reports of AB (p.111), Rotterdamsche Bankvereeniging (p.228) and Incasso Bank (p.137) in Van Oss' Effectenboek 1915-1916 Binnenland. Twentsche Bank outflows are computed based on monthly balance sheets published in newspapers (accessed via Delpher).

[19] The agent of the Enschede branch writes in the 1914-1915 report to the head office that "More than ever before have current-account holders used the Nederlandsche Bank" and depositors have "temporarily foregone the benefits of interest remuneration on their deposits at other banks". The agent in Deventer writes that "Many withdrew their money from deposit-taking institutions and deposited in current account at DNB". A branch close to the war border reported that "several individuals have deposited substantial amounts at the Bank, since they lost trust in other bankers or institutions, despite foregoing interest on their accounts." NA, 2.25.68, 706: Enschede, 1914-1915, p.5; 137: Deventer, 1914-1915, p.13; 104: Arnhem 1914-1915, p.14, respectively.

Bank, 1915, p.3.). The most illustrative example comes from the Maastricht branch of DNB:

"Persons, who previously would not have thought of opening an account at the Nederlandsche Bank, have foregone interest in favor of certainty to unconditionally be able to have their money at their disposal."[20]

Virtually all commercial banks raised their deposit rates in the first days of August to combat the outflow of deposits. They advertised their higher deposit rates in both national and regional newspapers (see Figure B.4 for an example). Some went further and published announcements to try and calm depositors. For example, the Utrechtsche Bankvereeniging published in the regional newspaper on 23 August:

"Following rumors, that bankers would have refused to pay out deposits, we declare, that ALL of the demandable and matured deposits have since the start of the crisis been DIRECTLY PAID OUT BY US."[21]

Other banks even published joint statements in regional newspapers that they would only pay out the amounts "necessary for living costs" or suspended payments for a month.[22] One DNB branch reported that it had to assist one of the commercial banks to counteract the outflow of deposits after the one-month suspension of payments. This anecdotal evidence suggests that banks experiencing a run borrowed more from the central bank, at a higher rate.[23]

The deposit withdrawals during the bank run were temporary: the largest commercial banks mentioned earlier had higher end-of-year deposit balances in 1914 compared to 1913.[24] Similarly, several central bank branches mention in their yearly report over 1914 that many of the retail deposits have subsequently been withdrawn again to find "a more profitable way for investing their excess cash".[25] DNB started lowering its discount rate and other lending rates on August 20 when they perceived the most acute phase of the panic to be over (De Nederlandsche Bank, 1915). The total discount window lending reached its peak of $f.$ 186 million in November 1914. The stock market reopened on February 9th, 1915, providing another source of funding again for banks. Despite the temporary nature of the shock, the central bank retail deposit facility might have been a wake-up call for more exposed commercial banks.

[20]NA, 2.25.68, 206: Maastricht, 1914-1915, p.26, author's own translation.

[21]Utrechtsch Provinciaal en Stedelijk Dagblad, 23-08-1914, accessed via Delpher, author's own translation.

[22]Nieuwsblad van Friesland, 05-08-1914; Limburger Kourier, 01-08-1914, accessed via Delpher.

[23]NL-NaHa, 2.25.68, 206.

[24]Van Oss Binnenland 1915-1916 contains annual reports of banks mentioning that all withdrawals have returned (see Appendix A.3).

[25]NA, 2.25.68, 4: 's Gravenhage, 1914-1915, p.12, author's own translation.

2 Data

This paper draws on a combination of archival sources covering the period 1910-1918. The first source consists of historical newspapers from the years 1914-1915. Delpher is an online newspaper database consisting of more than 2 million newspapers from 1618 - 1995. The newspapers are both regional and national and include advertisements of Dutch commercial banks. These banks announced, with varying frequency, (change in) their deposit rates, balance sheets and opening of new offices. I hand-collect any deposit rate advertisements for 79 banks from 47 different newspapers. This yields deposit rates by maturity—demandable, weekly, monthly, quarterly, and annual—for 79 banks, of which 39 report both a demandable and a weekly rate. 76 banks have a short-term (demandable, daily or weekly) rate advertised and are included in the main sample.[26] For only a subsample of 14 banks, the newspapers contain monthly or quarterly deposit statements, allowing for higher-frequency analysis of deposit dynamics around the outbreak of WWI.

Figure B.4 is an example of an excerpt from one of the national newspapers, Algemeen Handelsblad on 1st of September 1914. As can be seen from the figure, commercial banks had their deposit rates advertisements often printed on the same page as their competitors. The advertisements included the deposit tenors (depending on which type of deposits the bank offered), the rate and often the locations of its branches. Two things stand out in the advertisements. First, there is substantial heterogeneity among banks in their deposit rates. For example, despite the bank runs having subdued in September 1914, the Credietvereeniging Amsterdam still pays 4% interest on demandable deposits, compared to three percent by the other banks. Second, the example shows that all banks except the Nederlandsch-Indische Handelsbank offer an equal or higher rate for a shorter deposit tenor. Credietvereeniging Amsterdam again stands out as it offers 4.5% for 1 month term deposit, while it offers 4% for a 3 month term deposit. This indicates that these banks especially needed liquidity in the short term. In the empirical analysis I use the inconsistent deposit pricing, or inverted term structure of deposits, as an outcome variable to proxy for financial distress.

The second source is the archive of De Nederlandsche Bank (DNB), available at the Dutch National Archive. The DNB historical records provides daily aggregate balance sheets, including the balance of current account balances and end of year lists of current account holders. Together, these records allow me to track the evolution of (non-)bank accounts at the central bank at both the aggregate and office level, and to distinguish between different types of depositors—individuals, firms, government bodies, and financial institutions.

[26] Of the 76 banks, 67 advertise both before and after the outbreak of the war. Four banks do not advertise before the war, and 5 banks do not advertise after the war. The DiD estimate thus determined by 67 banks.

Third, the paper uses digitized and standardized commercial banks' balance sheets for 75 depositing banks from 1910-1930 from several sources: Van Oss Effectenboek, depositories at the DNB archive and commercial banks' individual archives.[27] The sample in the regression analysis runs from 1910-1918, as the effects of central bank exposure will likely be conflated with post-war deflation and the Dutch banking crisis in the 1920s.

Fourth, I use the 1914 Financieel Adresboek (Financial Address book) to identify the locations of all banks and financial institutions that appear among DNB's current account holders. Moreover, this source is used to create a local competition measure based on the number of financial institutions active in a city. This financial yearbook contains all (self-reported) financial players, from deposit banks and brokers, to mortgage banks and insurance agents. The book reports for each bank and firm the founding year, its locations, its representatives, whether it has access to a stock exchange (you needed to be a member to be allowed to trade in the building), and for public entities the names of the directors. I manually search for matches with the DNB current account holder lists and create a dummy that equals one if a person or institutions appears in the book. The address information is essential for constructing the geographic proximity measure central to the identification strategy.

Finally, to obtain pre-shock regional characteristics, I collect data of the 1909 census ("*Beroepstelling*"). While the census is very granular in terms of coverage of occupations and geographical coverage (municipal level), it unfortunately does not include regional GDP. It provides total population of each municipality and the unemployment in each region.

2.1 Summary Statistics

Figure 3 shows the evolution of commercial bank average deposit rates with different tenors from 1913-1915. The figure looks similar for median deposit rates. I collected any tenor that banks advertised but standardize them into 6 categories: The demandable deposit rates (in red) include 0 and 1 day maturity, weekly deposit rates have a tenor between 8-15 days, monthly rates have a monthly tenor, quarterly rates are 2 and 3 months combined, and semi-annual and annual are 6 and 12 months tenors, respectively. Not surprisingly, the deposit rates for longer-tenor deposits are on average higher than shorter deposits. One exception to this is the demandable deposit rate around the outbreak of the war: on average the demandable deposit rate reaches a similar level to the monthly rate.

[27] I thank Remo Oostdam for the collaboration in constructing the standardized Dutch bank-level dataset used in this paper.

The longer deposit tenors are nearly flat and identical for virtually all banks. In general, the longer the maturity, the fewer variation in the average deposit rate: the annual rate is consistently around 4%. Table 1 shows the smallest standard deviation for the annual rate (0.23), and figure 3 shows little variation in the deposit tenors above 3 months. It thus appears that deposit competition via rates was mostly done on the shortest deposit tenors. Moreover, the shortest deposit tenors are likely the closest substitute to central bank retail deposits, which had no remuneration and were demandable.

Bank deposit rates should be put into perspective comparing them to alternative sources of bank funding. Figure 4 plots the evolution of the shortest-tenor deposit rate (in red) with the money market repo rate (in grey) and the central bank discount window borrowing rate. While short-term bank deposit rates reach their highest levels after the shock, deposits are still the cheapest source of funding for commercial banks. DNB's discount window rate shortly stood at 6% before remaining at 5% far into the summer of 1915. Importantly, the money market rate was kept artificially high at 7% (6%) due to the closure of the Amsterdam Stock Exchange. The most likely alternative to deposit funding was thus borrowing at the discount window of the central bank. Indeed, once the stock market reopens on February 9, 1915, the money market rate quickly drops again, and the deposit rate follows.

Table 1 reports the summary statistics of the deposit rate sample. The number of banks varies with each variable due to 1) data availability, 2) the type of deposit rates each bank offers and 3) non-perfect overlap of the banks from each source. In the regression analysis, the results will always contain a minimum of 39 banks. What stands out from Table 1 is that the demandable, short-term and weekly rate are very similar in terms of distribution. To maximize the sample size, I will therefore use the 'short-term' rate for the main analysis. The short-term rate is the rate that each bank offers for its shortest deposit tenor available. To ensure comparability, the tenor cannot be longer than 15 days to be included in this variable.

On average, commercial banks had close to three branches across the Netherlands. These branches covered most areas of the Netherlands and did not solely operate in the central bank cities (see Figure 5). The overlap between commercial bank and central bank branches was on average 74%, meaning that banks on average had three quarters of their business in the same location as a safe haven alternative. This relies on the simplistic assumption that all branches were equally large in terms of their deposit ratios, which due to data availability is impossible to test. High central bank (CB) exposure is a dummy that equals 1 if a commercial banks has all its branches in the same location as central bank offices. This is the case for 61% of the banks in the sample.

3 Empirical Approach

As a first stage, I analyze the causal effect of a central bank depositing alternative on the deposit-rate setting behavior of commercial banks. That is, I use a difference-in-differences setting with the share of a commercial bank's branches located in a central bank city as the treatment. The identifying assumption is that banks located near and far from a central bank office would have followed similar trends in their deposit rates and funding conditions absent the outbreak of the war. Moreover, it is key for my setting that depositors deposit their money at a bank in their (nearest) city. Across 1913-1914, 78% of account holders were located in the same city as the central bank branch where they deposited. Figure 6 show the geographical spread of account holders before and after the outbreak of the war, respectively. The number of retail depositors increased at almost every central bank office in 1914, indicating that across the country, an account at DNB became attractive only once depositors were seeking a safer place to deposit their money.

The identification strategy relies thus on the assumption that the geographic proximity between commercial and central bank offices is orthogonal to unobserved determinants of changes in deposit competition. This assumption is plausible for three reasons. First, DNB established its branch network in 1864 with minor changes until 1903, decades before 1914, primarily to facilitate discount and payment operations rather than deposit-taking. Second, commercial banks selected their locations based on local economic conditions, not in response to a central bank deposit facility that was, in practice, dormant. The central bank discount facility was also available at the smaller regional correspondent offices, which did not have the deposit facility. Third, the sudden deposit flight triggered by the outbreak of WWI was unanticipated by both commercial banks and the central bank.

Table 2 compares observations of banks with high (full) and low geographical exposure to central bank offices before the outbreak of the war. By design, the continuous measure of central bank exposure is higher for high-exposed banks. The overlap of bank branches with central bank offices for low-exposed banks is on average 28%. While for all deposit tenors, the rate offered by high-exposed banks is on average slightly lower, this difference is not statistically significant. Not surprisingly, low-exposed banks have more than twice the number of branches than high-exposed banks, and this difference is significant. Almost mechanically, more branches lead to lower overlap with the central bank and thus lower CB exposure. In the regression analysis the number of branches is included as a control variable.

For the balance sheet characteristics pre-war, the data is only available for 42 banks of the rates sample. Importantly, the high- and low- exposed banks are on average comparable in size. High-exposed banks have ex ante slightly more deposits and capital than low-exposed banks. Finally, low-exposed banks are more often located close to the war border (though this difference is not significant), which could affect run dynamics during the war. Only secured funding differs before the shock: high-exposed banks have on average 15% of funding secured, compared to 25% for low-exposed banks (significant at 10%). Since secured funding consists of repo borrowing, which is completely shut down once the stock market closes, and bills of exchange discounted at the central bank, it is unlikely that the difference in this variable can explain the variation in deposit pricing after the shock. If anything, one would expect banks relying more on secured borrowing to be more at risk after the war outbreak: repo borrowing is illiquid and central bank borrowing very expensive.

The DiD specification is as follows:

$$Y_{i,t} = \beta_1 CB\ Exposure_i \times Post_t + \Gamma_1' X_{i,1909} \times Post_t + \Gamma_2' X_{i,1913} \times Post_t + \kappa_i + \eta_t + \lambda_{g,t} + \epsilon_{i,t} \quad (1)$$

Where $Y_{i,t}$ is either the deposit rate, or a dummy for an inverted term structure of bank i at week t. $CB\ Exposure_i$ is the treatment intensity of bank i: the fraction of bank i's offices located in a CB city. $Post_t$ is a dummy equal to 1 for all weeks from 27 July 1914 onward. $X_{i,1909}$ is a vector of control variables on the municipality level from the 1909 census (i.e. before the shock), such as log(population) and the unemployment rate. $X_{i,1913}$ is a vector of control variables on the bank-level, such the competition for bank i (measured as the average number of competing banks for bank i, population-weighted over its branches) and importantly, the number of branches of a bank before the shock.[28]

The number of branches a bank has before the shock matters for three reasons. First, CB exposure is a *share*, so it is mechanically related to the geography of a bank's network. A single-office bank located in a central bank city takes the value one by construction, whereas a bank with offices spread across towns with and without a DNB office cannot. By design, mean CB exposure falls monotonically in network size, from 0.86 among single-office banks to 0.33 among banks with four offices, and 36 of the 43 fully exposed banks in the rate sample operate a single office. Absent the branch network control, β_1 would conflate the effect of central bank proximity with the effect of operating a concentrated network. Second, it serves as a proxy for bank size. Large banks may have responded differently to the August 1914 liquidity shock for reasons unrelated to

[28]Both vectors are interacted with the $Post_t$ dummy since they are time-invariant and would otherwise be absorbed by the bank fixed effects.

central bank proximity: wider access to interbank funding, greater name recognition among depositors, and a more diversified deposit base. Third, a multi-branch bank can reallocate liquidity internally across its offices when a single location comes under pressure (Quincy, 2024). Such an internal capital market is itself a partial substitute for access to the discount window, so network size proxies for a competing liquidity technology rather than for scale alone.

Conditioning on $\text{Branches}_i \times \text{Post}_t$ sharpens the interpretation of β: it compares banks operating the *same number of offices* that differ in the share of those offices located in central bank cities.[29] Because both variables are determined by branch-location decisions taken well before 1914 and are fixed within bank throughout the sample, the control is pre-determined with respect to the shock.

I include bank (κ_i), week (η_t) and region-time ($\lambda_{g,t}$) fixed effects to account for unobserved heterogeneity in bank specific circumstances, war-related developments that vary by week and regional differences that might affect the deposit business of banks. Region fixed effects are defined as either 4 regions of the Netherlands (north, middle, south and centre, or 'Randstad') or the 20 ressorts that the central bank offices were divided into: see Figure B.1. The regional fixed effects are preferably interacted with monthly rather than weekly fixed effects to preserve sufficient within-group variation.[30] Standard errors are clustered on the commercial bank level.

Second, I look at bank balance sheet outcomes of high-and low exposed banks. Balance sheets are available on the yearly level. Similar to equation 1, I then regress several bank funding and lending variables on central bank exposure. In this case, $Y_{i,t}$ is either a funding composition variable (deposits, (un)secured funding, total equity) a profitability measure (net interest margin), or a lending variable (total lending, liquid and illiquid lending, securities lending) or a measure for liquidity (liquid assets). All outcome variables are weighted by total assets pre-war, using 1913 total assets as denominator. Post is now a dummy that equals one if the period is 1914 or later. 1914 is included in the post period as banks report their balance sheets end of each year. The time fixed effects are then yearly. The balance-sheet regressions control for branch network size $\times$ Post and the logarithm of the total yearly population of the bank's pre-war office locations. Unemployment is only observed in 1909 and the number of competing banks in 1914; Tables A.10–A.11 add them.

[29] Omitting the control biases β *downward*. Banks with larger networks raised rates more ($\hat{\gamma} = 0.049$, s.e. 0.014), and network size is negatively correlated with CB exposure (bank-level correlation -0.265), see Table 2.

[30] Using region x week fixed effects does not alter the coefficients substantially, but might reduce the statistical power in some specifications. Appendix Table C.8 shows that the coefficients are very stable irrespective of the fixed effects used.

Table 9 shows the balancing table using annual banks' balance sheets. Commercial banks in this sample relied for about 30% on deposits in their funding mix, and this is comparable for both high- and low exposed banks. This relatively low number compared to deposit-taking banks nowadays – around 55% in the EU and 75% in the US[31] – can be explained by the higher capital buffers that banks at the time voluntarily held: the average bank had 27% of its liabilities in capital and retained earnings. The findings of this paper can thus be seen as a lower bound: the response to central bank competition would likely be even stronger if banks rely more on deposits. The repo market was seen as a liquid and safe investment before the closure of the stock market. Banks both engaged in lending and borrowing in the repo market, with repo lending making up 12% of total assets.

4 Results

This section presents the empirical results in three phases: short, medium and long-term. The short term covers the crisis phase, which is the period right after the outbreak of the War (July - August 1914). The medium term focuses on the period after the initial panic and until the reopening of the Amsterdam Stock Exchange (September 1914 - February 1915). The long term focuses on the years during the War (1914-1918).

4.1 Crisis phase

4.1.1 Commercial banks' deposit rates

Figure 10 shows simple averages of short-term deposit rate of high- and low-exposed banks from 1914-1915. Before the outbreak of WWI, deposit rates are very similar for banks located in central bank cities and banks located elsewhere. In the months before the war, both the treatment and control group lower their rates. High-exposed banks even offer a lower rate in the weeks before the 28th of July, albeit the difference is less than 0.2 percentage points. Importantly, the high-exposed banks throughout the pre period appear to offer similar or slightly lower rates compared to the low-exposed banks. Once the war breaks out and triggers a bank run, the high-exposed banks increase their deposit rates instantaneous by a full percentage point. Low-exposed banks only increase the short-term rate by about half a percentage point. The high-exposed banks continue

[31] EBA Risk Assessment Report, June 2025:
https://www.eba.europa.eu/publications-and-media/publications/liabilities-funding-and-liquidity-1;
FED Statistical Release Asset and Liabilities of Commercial Banks in the United States, 13-02-2026:
https://www.federalreserve.gov/releases/h8/current/h8.pdf, respectively.

to offer slightly higher deposit rates than their low-exposed peers until the reopening of the stock exchange in February 1915, which likely provide them with another funding alternative again.

To test whether the difference is statistically significant, I run the DiD specification 1. Table 3 shows that after the war outbreak, demandable deposit rates increased 44 basis points across all banks. However, banks more exposed to central bank deposit competition raise their short-term deposit rates more aggressively following the outbreak of the war. The effect is concentrated in the acute crisis period. For the demandable deposit rate, the interaction between CB exposure and the post-war dummy is insignificant over the full 1914–1915 period, turns positive for 1914 alone, and becomes significant when restricting the sample to the crisis months July-August 1914. The magnitude increases the more narrow the time window: for July and August, banks more exposed to the central bank increased their deposit rates an additional 46 basis points. The coefficient for the continuous treatment effect increases from an insignificant –0.127 over the full period to 0.202 within 1914, to 0.458 in the July-August window. Since the variable CB exposure measures the fraction of a bank's branches located in a central bank city, these estimates imply that a fully exposed bank raises its demandable deposit rate by approximately 0.46 percentage points (or 46 basis points) more than an completely unexposed bank during the peak of the crisis. Compared to an averaged demandable deposit rate pre-war of 2.88 %, this is a large effect. When controlling for the number of offices of the commercial banks pre-war, the coefficient increases to 54 basis points. That is, conditional on network size, a fully exposed bank increases it's demandable deposit rate by an additional 54 basis points compared to a non-exposed bank.

Figure 7 plots the weekly coefficients over the full period 1914-1915. The figure can be split into three periods: one before the outbreak of the War (January-27 July 1914), the war period when the Amsterdam Stock Exchange is closed (28th of July - 8th of February 1915) and the war period when the stock exchange is reopened (9th of February - 31 of December 1915). Note that that the specification underlying this plot uses region $\times$ week fixed effects (on top of bank and week fixed effects) to absorb any regional unobserved characteristics that could influence banks' rate setting behavior. The difference between the rates of high- and low exposed banks before the outbreak of the War is small and not statistically significant, in line with the balanced-ness pre-war as shown in Table 2. Once the war breaks out, the high-exposed banks charge significantly higher demandable deposit rates for three weeks. Even though the coefficients remain large and positive after that period, there is substantial variation in the rate setting across high-exposed banks. After the reopening of the stock market, the average rate differential declines gradually, until the high-exposed banks offer slightly lower rates compared to low-exposed banks

(though this difference is not statistically significant). The average difference between high-and low exposed banks is zero in the period after the reopening of the stock exchange.

Figure 8 confirms the strongest effects in the crisis period. The figure shows the weekly coefficients for the shortest deposit rate offered by the banks. In the ten weeks (starting mid-June) before the war, the difference in deposit rates between high- and low-exposed banks is indifferent from zero. In the week of the war outbreak and the Dutch mobilization, the difference becomes positive and significant. Despite substantial standard errors, the effect remains significant throughout August, when the deposit rates gradually decline again. On average, the difference between demandable deposit rates are of high-exposed banks compared to low-exposed is higher than in the ten weeks before the shock, but after the significance disappears after three weeks. The results are robust when including additional fixed effects or controls (see Appendix Figure B.6).

Since the central bank retail deposits are demandable and do not offer remuneration, one might expect that the results should fade out over the maturity of the deposits. Indeed, Table 5 shows that the effects are concentrated on the shortest tenors (demandable and weekly), which are the most comparable to the central bank retail deposit accounts. Table 5 shows that the coefficient remains positive and large for all tenors (though not statistically significant for maturities above one week), but declines in magnitude the longer the tenor. These results suggest that only the deposits most sensitive to central bank competition were higher remunerated after the shock.

Banks that are more exposed to the central bank also are more likely to adopt an inverted term structure of their deposit pricing. That is, high-exposed banks more frequently offer a weekly rate above a monthly rate, or a demandable rate that is higher than the rate on weekly or monthly tenors. Table 6 shows that, during the crisis months, high-exposed banks are 20 percentage points more likely to adopt such an inverted deposit pricing scheme. Pre-war, the average probability of pricing the deposits inconsistent with their tenor was around 5%, so the effect is more than three times the pre-war mean. After the shock, fully-exposed banks became four times as likely compared to non-exposed to adopt an inverted pricing structure. The inverted term structure of a bank's deposit pricing suggests that this bank was especially liquidity constrained and in need of short-term funding.

4.1.2 Regional variation

One concern could be that the results are driven by exposure to large, or in other ways, important cities. This section discusses several specifications that show that the exposure to the central bank remains relevant when controlling for economic conditions, market

structure of the local deposit market. The point estimates are also stable when excluding the largest cities from the sample, but the reduced samples limit precision in the narrowest window.

4.1.3 Municipal characteristics

The regional controls city size (population) and unemployment describe the local markets a bank operates in rather than the bank itself. Each is measured at the municipality level and then aggregated over the bank's office locations, using the same office denominator that defines $CB\ Exposure_i$. Because both are measured once before the shock, their levels are absorbed by the bank fixed effects and they enter only interacted with $Post_t$.

Column (1) of Table 4 omits the network-size control and yields $\hat{\beta} = 0.210$ (s.e. 0.139), indistinguishable from zero. Adding $\text{Branches}_i \times \text{Post}_t$ in column (2) raises the estimate to 0.372, significant at5%, and the remaining regional controls leave it essentially unchanged. The same pattern holds in the July to August window, where the estimate rises from 0.372 to 0.538. Columns (3) to (5) add the log of total population served, the unemployment rate and the number of banks in the bank's locations, and leave the estimate essentially unchanged at 0.409, 0.408 and 0.395 respectively, each significant at the one percent level. None of the three municipal controls is individually significant over 1914: population served enters with a small negative coefficient (-0.076 , s.e. 0.075 in column (5)) and the unemployment rate with a coefficient close to zero. What the local economy looked like therefore does little to explain which banks repriced; what matters is how many offices a bank had and how many of them sat next to a central bank.

The July to August window is less uniform. The estimate rises from 0.372 in column (6) to 0.538 once branch network size is included in column (7), but falls back to 0.340 when population served is added in column (8), and remains at 0.350 in columns (9) and (10). Population served is itself significant in this window (0.111, s.e. 0.055): in the six weeks around the shock, banks serving larger markets raised their rates more. The two effects are not cleanly separable at this sample size — the difference between 0.538 and 0.340 is roughly one standard error — but the central bank coefficient remains positive and significant at the ten percent level throughout, and the point estimate stays above the 0.372 of the uncontrolled specification in column (6).

A finer geography does not overturn this. Table C.8 replaces the region–month fixed effects with region–week fixed effects, so that the comparison is between banks in the same region in the same week. With the full set of controls the estimate is 0.425 (s.e. 0.130) over 1914 and 0.471 (s.e. 0.185) in the July to August window, both larger than under region–month fixed effects.

4.1.4 Deposit market structure

A second possibility is that the result reflects the competitive structure of the local deposit market rather than proximity to the central bank: banks in crowded markets may have to bid more aggressively for deposits, and crowded markets are also where central bank offices are. I measure local competition as a count of all deposit-taking institutions in a city, and the variable is the population-weighted average of this count across a bank's office locations.

Column (5) of Table 4 adds # Banks$_i$ $\times$ Post$_t$. The coefficient is 0.004 (s.e. 0.005) over 1914 and 0.000 (s.e. 0.007) in the July to August window, and the central bank exposure estimate is unaffected in both. The number of competing deposit-takers in a bank's markets thus has no detectable effect on how much it raised rates after the shock, and does not absorb the central bank effect. This is perhaps the cleanest indication that the mechanism is not generic competitive pressure: a bank in a town with fifteen other banks and no central bank office behaved no differently from a bank in a town with three.

4.1.5 Large vs. small cities

The composition of the local economy is a related concern, since a bank's depositors differ across an agricultural town and an industrial one. Table 4 uses only the banks that are comparable in terms of city size. The four largest Dutch cities (Amsterdam, Rotterdam, The Hague and Utrecht, each above 100,000 inhabitants) all have a central bank office and have no counterfactual: there is no comparably sized Dutch city without a central bank office. If the estimated effect is really a big-city effect, it should disappear once those cities are removed.

I trim the sample to the common support in log average market population between highly and less exposed banks, which retains 41 banks whose local markets are comparable in size by construction. Columns (1) and (2) of Table 4 give 0.261 (s.e. 0.201) without controls and 0.345 (s.e. 0.165) with them, the latter significant at the five percent level and almost identical to the full-sample estimate. In the July to August window the trimmed sample contains only 349 observations and the estimates, 0.209 and 0.361, are stable in magnitude but no longer significant. Table C.5 addresses the city size dispersion in a different way, by excluding the largest 4 cities. The triple interaction is 0.079 (s.e. 0.678), close to zero and very imprecisely estimated, so there is no evidence that the repricing response was concentrated in the largest cities. Column (3) of Table C.5 instead drops every bank with any branch in a B4 city, which removes 41 of the 76 banks and more than half the observations. The estimate on the remaining 35 banks is 0.383 (s.e. 0.151), close to the full-sample estimate of 0.417 in column (2) and still significant at the five percent level.

In the July to August window the same exclusion leaves only 279 observations and the estimate falls to 0.220 (s.e. 0.156); the point estimate remains positive and of similar order, but the window is too short and the sample too small to sustain inference.

Columns (3), (4) and (7) to (9) of Table 4 replace the region–month fixed effects with fixed effects for the bank's dominant 1909 sector interacted with month, so that banks are compared only to other banks serving the same kind of local economy.[32] The estimate is 0.320 (s.e. 0.151) over 1914 and 0.454 (s.e. 0.148) in the July to August window. Columns (4) and (8) further drop agriculture, the one dominant sector that contains no highly exposed bank and therefore contributes no within-stratum comparison, leaving 0.317 and 0.478 on 67 banks. Adding the regional controls on top in column (9) gives 0.526 (s.e. 0.150).

Taken together, the size-based tests point the same way as the municipal controls. The effect survives dropping the four largest cities, and trimming to comparable market sizes, with point estimates varying in the 0.21 to 0.53 range against a full-sample baseline of roughly 0.42. Where it loses significance it does so in the narrow July to August window, where the trimmed samples fall to a few hundred observations, rather than in the specifications that retain power.

4.2 Return of deposits and reopening of the stock market

Deposits returned to the banks within the year. Figure 9 shows the evolution of deposits for banks that did report deposits at a higher frequency (either monthly or quarterly, published in newspapers). While it only covers a handful of banks, it provides suggestive evidence for two things. First, deposits returned to pre-war levels at the end of 1914. The coefficient plot Figure 11 confirms that with yearly data, the 1914 coefficient is not significantly different from the years before the shock. Second, the high-exposed banks that published their statements in newspapers seem to have experienced a larger outflow of deposits compared to low-exposed banks in the months after the shock.

The Amsterdam Stock Exchange reopened on February 9, 1915. From that moment onward, banks had an additional funding channel at their disposal again: collateralized borrowing via the repo market. Figure 4 shows that immediately after the reopening, the money market rate (which had been artificially kept high), dropped until the lowest value of the sample period in July 1915. The outstanding repos before the outbreak of the War had been frozen while the stock exchange was closed, and were only allowed to

[32]The dominant sector is the largest of eight non-residual 1909 occupational categories in the bank's population-weighted office average.

be settled after July 1st, 1915. Only in the summer of 1915 banks were thus completely unconstrained again in managing their balance sheet.

More exposed banks especially lowered their deposit rates after the reopening of the stock exchange. Figure 10 plots the average deposit rate across high- and low-exposed banks for the entire period. It suggests that the demandable deposit rate of high- and low-exposed banks was on average very similar before the reopening of the exchange. The rate for high-exposed banks declined faster and the gap between the high- and low-exposed banks is larger than before the shock from July 1915 onward.[33]

Table 7 splits the central bank exposure effect into three time periods: when the exchange is closed, when the stock exchange reopens, and when the repos are unfrozen. All interactions are relative to the pre-war rate differential between more and less exposed banks. Unsurprisingly, the coefficient of central bank exposure during the closure of the exchange is between 34 and 40 basis points, in line with the findings before (Table 3 and 4). Compared to pre-war, more exposed banks still offer a higher rate compared to low-exposed even when the exchange reopens. It is only after the settlement of the frozen repo transactions in the summer of 1915 that the coefficient becomes negative, though this difference is not statistically significant.

Banks that are larger (above median total assets in 1913) might have more alternative means of funding. Similarly, banks with an Amsterdam branch likely have better access to the money market. Indeed, Table 7 shows that both larger banks and banks with an Amsterdam branch significantly lower their deposit rates after the reopening of the stock exchange. While the number of banks in each group only allows for double (rather than triple) DiD, the results are consistent with access to alternative funding driving the rate cuts once markets reopened.

The medium-term results are consistent with banks exposed to outside competition moving away from deposit funding, rather than depositors moving away from the exposed banks. Deposits flow back at the end of the year and once alternative means of funding became available, exposed banks did no longer pay a premium to attract deposits.

4.3 Long-term effects on commercial banks' balance sheets

4.3.1 Bank funding

The natural follow-up question after commercial banks' raising their deposit rates is what happens to their funding structure, and ultimately, their lending portfolio. Due to the

[33] In the same week as the release of the frozen repos, DNB lowered its discount window rate by 50 basis points. While I cannot separate the two events, they both made non-deposit funding cheaper.

annual nature of banks' balance sheet variables, it is impossible to quantify the exact outflow of deposits at the time of the crisis. Despite the temporary additional increase in deposit rates, banks potentially reassessed the cost of deposit funding and adjust their funding mix accordingly afterwards. Especially high-exposed banks might hold a safer or more liquid portfolio to meet any future deposit outflows, or might rely more on other types of funding. Figure 11 shows that indeed, banks more exposed to the central bank competition for deposits experience slower deposit growth in the years after the outbreak of the war. In 1914, both high- and low-exposed banks have a similar level of deposits compared to pre-war, in line with the anecdotal evidence that at least the largest banks saw deposits at levels compared to the preceding year at the end of 1914. However, the years 1915-1918 show that the banks more exposed to the wake-up call of central bank competition in 1914, grow significantly slower than banks less exposed.

Table 10 shows that the high-exposed banks did on average slow their asset growth, which is mostly driven by the slower deposits growth. Deposits as a fraction of pre-war total assets grew 0.39 less compared to non-exposed banks. Since virtually all banks were growing in normal terms during the war, the result can be put into perspective by looking at the pre-and post-shock mean. On average, commercial banks doubled in deposits in terms of 1913 assets (growth 0.634 - 0.303 = 0.331). The high-exposed banks growth coefficient is 0.39, implying that their growth is about a third of low-exposed banks. Table C.9 divides each liability by current total assets. The deposit share of more exposed banks falls by 9 percentage points relative to a pre-war mean of 31%, while total assets do not shrink significantly: exposed banks shifted their funding mix away from deposits rather than merely growing more slowly. The change in funding composition is not being offset by additional secured or unsecured funding. Table C.10 shows that the effects are larger and more precise among the 51 banks with comparable market size.

4.3.2 Bank lending

Since banks that rely less on deposit funding have likely seen an increase in their funding costs (see Figure 4), they may have altered their lending portfolio accordingly. Since the data does not allow to control for within-borrower (Khwaja and Mian, 2008) or within-industry fixed effects (Degryse et al., 2019), despite controlling for local characteristics of bank's branch location, the results on bank lending should be seen as suggestive. Table 11 provides suggestive evidence that the high-exposed commercial banks grow slower in their lending, in about the same magnitude as their slower deposit growth. The decomposition of the total lending into liquid, illiquid and repo (or marketable) lending shows that there is a lot of variation between banks in altering their lending portfolio. Surprisingly, more exposed banks do not hold significantly more liquid assets after the shock.

4.3.3 Small vs. large banks

Since the outbreak of war represents an aggregate shock unrelated to bank fundamentals, there is no obvious reason why total bank size (as opposed to a bank's branch network) would affect deposit rate responses to DNB competition independently of exposure intensity. Consistent with this, bank size is weakly correlated with treatment intensity in this sample (correlation = 0.053 for the binary measure, 0.225 for the continuous measure; see Table 2 for the equal average bank size for high- and low-exposed banks), suggesting size does not capture additional variation beyond what the treatment variable central bank exposure already measures. While previous literature has documented size-related differences in deposit rate setting in other contexts (Begenau and Stafford, 2025; Drechsler et al., 2017; Anyfantaki et al., 2026), these findings reflect equilibrium pricing differences in normal times rather than responses to aggregate shocks.

If larger banks can more easily substitute lost deposits with alternative funding sources such as interbank borrowing or securities lending, the effect of deposit outflows on their lending portfolio may be uneven across bank size. It is potentially also easier for larger banks to issue additional capital or bonds, after the reopening of the stock exchange. For this reason, I run a triple DiD with a large bank dummy as a third interaction:

$$\begin{aligned} Y_{i,t} = \beta_1 CB\ Exposure_i \times Post_t + \beta_2 Large\ Bank_i \times Post_t \\ +\beta_3 CB\ Exposure_i \times Post_t \times Large\ Bank_i + \kappa_i + \eta_t + \lambda_{g,t} + \epsilon_{i,t} \end{aligned} \tag{2}$$

Where the outcome variable $Y_{i,t}$ is again one of the bank balance sheet items. $LargeBank_i$ is a dummy that equals one if a bank is above median size in 1913, and 0 otherwise. The same set of fixed effects apply as before and standard errors are clustered on the commercial bank level.

Table 12 shows that there is little evidence for heterogeneity in bank size driving the results. The deposits coefficient in column (2) is significant for the main interaction variable of central bank exposure and post, which now reflect the coefficient for the small banks. However, the coefficient for the large banks ($CB\ Exposure_i \times Post_t \times Large\ Bank_i$) in column (2) is positive but insignificant. Due to a lack of power in smaller samples, the estimates are not precise enough to draw any conclusions on bank size potentially affecting the composition of its balance sheet.

4.4 Robustness

4.4.1 War border exposure

A concern for the identification is that the results are not driven by a nearby safer central bank retail deposit alternative, but by the direct threat of war. Banks located in provinces bordering belligerent countries may have faced stronger deposit outflows regardless of their proximity to a DNB office. To mitigate these concerns, I control for the war threat in Table C.2. I include a war exposure dummy that is equal to one if a bank has at least one branch in a war border province. Across all specifications in the narrow time window, the CB exposure interaction remains significant and of similar magnitude to the baseline estimates, while the war exposure interaction is generally insignificant and small. The results thus suggest that it is proximity to the central bank, not proximity to the war border, that drives the competitive response in deposit pricing.

4.4.2 Unit banks

The continuous treatment measure requires aggregating locations and aligning bank-level controls that are defined at the head office level, such as the region (ressort). Moreover, the continuous measure assumes that deposits are spread equally over the branches. To show that these assumptions do not affect the findings, I restrict the sample to banks that have only one branch. These so-called unit-banks have only binary exposure (either they are located in a central bank city or not) and the treatment effects are for that reason expected to bite most: those banks cannot rely on other, non-exposed branch locations to attract depositors. Unit banks have the additional benefit that they price their deposits for only one local market. Table C.6 shows that the number of branches matters for the demandable deposit rate. When restricting the sample to only unit banks, the exposed commercial banks increase their demandable deposit by an additional 70 basis points after the shock. Not surprisingly, the effect for unit banks is much larger and stronger than for multi-branch banks, who have the possibility to attract deposit funding in locations without a central bank office. The multi-branch banks also increase their demandable deposit rates, yet the effect is much smaller and not significant. The probability to price deposits in an inverted manner is of the same magnitude for both unit and multi-branch banks that are exposed to the central bank, both significant at the 5% level.

4.4.3 Stock market closure

Despite including regional (CB ressorts) fixed effects, a confounding factor could be the size of a city. Despite parallel trends before the outbreak of the war, larger cities might

see different run dynamics compared to smaller cities. Since the central bank exposure measure is correlated with city size (most large cities have a central bank office), it is difficult to disentangle the two. One main concern is that the closure of the Amsterdam Stock Exchange has different effects on banks exposed to Amsterdam, which might be driving the results. Table C.4 shows that when excluding the banks that have any branch located in Amsterdam, the results still hold and are of similar magnitude as before.

4.4.4 Binary treatment

In the baseline specification, I use a continuous treatment variable measuring the fraction of a bank's branches located in a central bank city. This is the preferred measure, as it exploits variation in the intensity of exposure within banks: an additional branch in a central bank city marginally increases a bank's exposure to deposit competition, comparing otherwise similar banks with slightly different geographic footprints. As a robustness check, I also estimate the regression using above median exposure (i.e. fully exposed, meaning all of a bank's branches are located in central bank cities) to test the effects of central bank competition. Table C.7 shows that the coefficients for deposit rates are slightly smaller but still positive and significant in the narrow time window. The inverted term deposit pricing effect is still positive but a lot smaller as well, and none of the results are statistically significant. This indicates that the 0-1 measure is likely too crude, since all banks that have less than full overlap are classified as non-exposed. It appears that even one branch overlap with a central bank office can affect deposit-rate setting.

5 Conceptual Framework

This section provides a framework to reconcile the findings that more exposed banks increase their demandable deposit rate more in the short term, but rely less on deposits in the long term. The point of this exercise is to disentangle depositor demand from bank supply of deposits.

Set-up. Consider unit banks that are price-setters for deposits in their local market. For historical idiosyncratic reasons, some cities have a central bank office that accepts retail deposits and other cities do not. There are three time periods: $t = 0$ (before the war), $t = 1$ (crisis at the outbreak of the war), and $t = 2$ (after the crisis).

Banks. There are two types of banks: banks that are located in cities with a central bank office (H) and banks located in cities without a central bank office (L). The banks are identical otherwise. Each bank operates under monopolistic competition in its local

deposit market, as in Drechsler, Savov and Schnabl (2017). Banks must raise their deposit rates to attract more deposits.[34] Each bank $j \in \{H, L\}$ maximizes its profit function:

$$\begin{aligned} \Pi_{j,t} = & \underbrace{\left(r_t + l_0 - \tfrac{l_1}{2} A_{j,t}\right) A_{j,t}}_{\text{asset return}} - \underbrace{\left(r_t + \tfrac{h}{2} W_{j,t}\right) W_{j,t}}_{\text{wholesale cost}} - \underbrace{\left(r_{j,t} + c_{j,t}\right) D_{j,t}}_{\text{deposit cost}} \\ & \text{s.t.} \quad A_{j,t} = D_{j,t} + W_{j,t} \end{aligned} \tag{3}$$

Similar to Drechsler, Savov and Schnabl (2017), the bank's problem has a quadratic cost function. Banks earn a return on their assets A that falls as the bank grows ($l_1 > 0$). Banks fund assets A with wholesale funding W, which cost $r_t + (\frac{h}{2} W)$ and retail deposits D, which cost $r_{j,t} + c_{j,t}$.[35] The market rate r_t is common to all banks.[36] The parameter $h > 0$ and reflects the speed at which W becomes more expensive, either due to collateral scarcity or outstanding loans in the interbank market.

Deposit demand curve. Banks face the following linear deposit demand:

$$D_{j,t} = \bar{D}[1 + b(r_{j,t} - v_{j,t})] \tag{4}$$

Where $\bar{D}$ is the depositor base at $t = 0$, which is independent of the bank's type.[37] The demand slope b reflects how flighty depositors at bank j are.[38] Bank j sets the demandable deposit rate $r_{j,t}$ at each period t. Demand for deposits at bank j also depends on the outside option for depositors $v_{j,t}$, which is determined by the location of bank j and defined in equation (7).

Deposit rate setting. The bank chooses its optimal deposit rate $r^*_{j,t}$ each period:

[34] To keep the framework simple, I assume the market power is the same across banks, such that H and L banks are identical. In the empirical analysis I control for differences in local competition.

[35] The costs per deposit guilder (dollar) can be seen as operating costs of the bank's franchise as in Drechsler, Savov and Schnabl (2021). It can include advertising, salaries, maintaining the branches etc. In the baseline of the simple model here it is the same for all banks and periods, i.e. $c_{j,t} = c$.

[36] The market rate is the money market (*prolongatiemarkt*) rate, which is common to all banks. During the closure of the stock exchange from 29 July 1914 – 8 February 1915, the money market did not operate and the only marginal source of non-deposit funding was discounting at the central bank. This facility was available for all banks at a penalty rate. The central bank discount window rate was therefore the market rate at $t = 1$.

[37] In practice a bank's depositor base carries over from one period to the next. I hold it fixed so that the crisis has no mechanical lasting effect. If depositors who left in the crisis do not return, a bank starts $t = 2$ with a smaller base.

[38] The slope b is common to H and L. If relaxed and using bank-specific $b_j > 0$ instead, all sign predictions are unchanged; b_j only scales the size of the responses, and its level effect is absorbed by bank fixed effects.

$$r^*_{j,t} = \tfrac{1}{2}\big(m_{j,t} + v_{j,t}\big) \tag{5}$$

Such that the bank's offered deposit rate depends on both the marginal value of deposits $m_{j,t}$ and the outside option for depositors $v_{j,t}$. The factor $\frac{1}{2}$ implies that while a higher rate attracts more depositors, the cost for existing depositors also increases.[39] With linear deposit demand, the bank sets the rate midway between m and v.[40] Equation (5) is at the core of the paper as it allows for distinguishing 1) banks supply of deposits from 2) depositors demand for deposits.

First, $m_{j,t}$ is the *bank-driven channel* of deposit rate setting: the higher the worth of a deposit guilder, the higher the rate $r_{j,t}$ the bank will want to offer to depositors. It reflects how much a deposit guilder net of costs is worth to the bank:

$$m_{j,t} = r_t + l_0 - l_1 A_{j,t} - c_{j,t} \tag{6}$$

The marginal value depends on the prevailing market rate r_t and the size of balance sheet A, which in turn depends on chosen amount of non-deposit funding W^* (balance sheet constraint). The cost of non-deposit funding differs by period, and $m_{j,t}$ is therefore evaluated at the bank's optimal balance sheet for that period. Each bank borrows until the marginal asset return equals the marginal wholesale costs $r_t + hW^*_{j,t}$. Similar to Drechsler, Savov and Schnabl (2021) and Drechsler, Savov, Schnabl and Wang (2026), higher operating cost per deposit guilder $c_{j,t}$ lowers the franchise per guilder.[41]

Second, $v_{j,t}$ is the depositors' outside option, or the *depositor-driven channel*: the higher the value of outside options, the higher the rate the bank needs to set to prevent outflows. A depositor i keeps funds at bank j if the rate offered is higher than the outside option ($r_{j,t} \geq v_{i,j,t}$), where the outside option is defined as follows:

$$v_{i,j,t} = v_t + e_j\,\Delta_t - \xi_i, \qquad e_H = 1,\ e_L = 0 \tag{7}$$

Each depositor has a personal attachment or preference for bank j, ξ_i, which is either convenience, relationship, habit or any other unobserved characteristic. I assume ξ_i is

[39] The factor $\frac{1}{2}$ follows from linear demand and a single price-setting bank; because $m_{j,t}$ falls with balance-sheet size, the effective pass-through of $v_{j,t}$ lies between $\frac{1}{2}$ and 1 (Appendix D), and only the sign of the $H - L$ difference matters for the predictions. The framework abstracts from competition between banks within a city; controlling for the number of banks leaves the results unchanged (Table 4).

[40] To be precise, the equation should be $r^*_{j,t} = \frac{1}{2}\big(m_{j,t} + v_{j,t}\big) - \frac{1}{2b}$. The term $\frac{1}{2b}$ reflects the market power of the bank, but since b is identical for H and L and constant over time, the markdown does not affect the difference in rates between the two types.

[41] The margin is $m_{j,t} - r_{j,t}$. When the asset return equals the market rate, $r_t - r_{j,t} - c_{j,t}$ is the franchise per guilder.

uniformly distributed and its distribution is constant over time and the same for bank H and L. Aggregating over depositors gives the linear demand in equation (4), with slope b common to both banks, where $v_{j,t} = v_t + e_j\Delta_t$ is the common part of the outside option. The exposure indicator e_j is binary and drops out for the L-type.[42] The common outside option v_t is the value of holding cash, which varies over time but not across banks. The H-type bank thus faces an extra option Δ_t, which varies over time.

Before the war ($t = 0$). Central bank retail deposits existed before the outbreak of World War I, but were rarely used. Commercial banks were perceived safe and a central bank retail deposit was not remunerated, thus the value of the extra outside option is negligible ($\Delta_0 \approx 0$). The deposit demand at bank H and L are subject to the same outside options and costs. This yields the testable implications that deposit rates and deposit volumes (as a fraction of total assets) are comparable for high- and low-exposed banks.

The crisis ($t = 1$). The outbreak of the war increases the value of safe alternatives for all depositors ($v_1 > v_0$). Moreover, the stock exchange closes which implies that wholesale funding is only available at the central bank's discount window, open to every bank at a penalty rate: $r_1 = r^{DW} > r_0$. Equation (5) shows that both factors force all banks to increase their deposit rates. The depositors of bank H have an additional outside option ($\Delta_1 > 0$) because of the location in a city with a central bank office: depositing at the central bank. The central bank exposure implies that bank H needs to increase its rate more than bank L.[43]

After the crisis, during the war ($t = 2$). Once the acute crisis phase is over, the value of the central bank retail deposit as outside option likely declines ($\Delta_2 \in [0, \Delta_1)$). Moreover, the stock exchange reopens, providing banks with cheaper alternative sources of funding (non-deposit funding is available at the market rate $r_2 < r_1$). Whether the crisis leaves a difference between bank H and bank L is an empirical question. The model distinguishes three cases:

1. Return to normal: bank H and bank L will set the same deposit rates, hold the same level of deposits and assets.

2. Shift in deposit demand: depositors might have persistently changed their valuation of the outside options ($\Delta_2 > 0$; Antipa and Rieder, 2026). In that case, bank H

[42] Banks in this paper set rates uniformly, so in the simplifying assumption of the model, unit banks have binary exposure $e_j \in \{0, 1\}$. In the continuous treatment used in the empirical analysis, think of the share of overlapping offices as $e_j \in [0, 1]$: a bank with multiple offices in different locations then faces the office-weighted average of their outside options.

[43] Following equation (5), the deposit rate rises by less than the outside option, and bank H will not be able to fully prevent outflows. The data used in the rest of the paper does not allow to track deposit flows at high frequency, and therefore the empirical analysis relies only on the rate response.

needs to offer a higher deposit rate than L, but will hold fewer deposits, as it cannot match the increased value of the outside option one-to-one.

3. Shift in deposit supply: banks might have persistently changed their perceived cost of deposit funding. During the war, all banks believe that the crisis may recur with probability $p > 0$, independent of their exposure to central bank offices.[44] The crisis in $t = 1$ showed that defending deposits is costly, especially for bank H. This raises the expected future costs for deposits more for bank H than for bank L ($c_{H,2} > c_{L,2}$).[45] Equation (5) then predicts that bank H pays a lower deposit rate and attracts fewer deposits.

In both the depositor-driven (2) and bank-driven (3) channel, bank H holds fewer deposits, uses more non-deposit funding and has fewer assets than L. The two cases differ only in the deposit rate: the rate moves in the opposite direction to deposits when demand shifts, and in the same direction when supply shifts (Drechsler et al., 2017). The sign of the long-run rate difference therefore reveals which force dominates. If both channels are simultaneously at work, a flat rate is possible, and the deposit rate alone cannot separate the two channels.

6 Conclusion

Potential alternatives for commercial bank deposits, such as stablecoins and central bank digital currencies, are increasingly being adopted. Competition for bank deposits affects banks' funding and lending decisions and has consequences for monetary policy pass through and credit provision. It is therefore crucial to understand the effects of competition for bank deposits. This paper uses the outbreak of World War I as an exogenous shock to show that during times of heightened uncertainty, a central bank retail deposit facility can trigger a flight-to-safety of retail deposits from commercial banks to the central bank. I use the overlap between locations of the central bank and commercial bank branches to define treatment intensity. Banks with higher exposure to the central bank raise their deposit rates more and are more likely to adopt an inverted term structure of deposit pricing following the outbreak of the war. This effect is concentrated in the months after the outbreak of the war and subsides afterwards. More exposed banks reassess the cost of deposit funding and have a lower share of deposits in their funding

[44] I cannot distinguish whether the channel is driven by rational learning or a salience shock. Both mechanisms are plausible and provide the same long term implications.

[45] Another explanation could be that the lending opportunities for bank H have declined, which also lowers the marginal value of an additional deposit guilder in equation (6). While I cannot provide causal evidence for lending effects, in the empirical analysis I show that banks especially reduced illiquid lending, which is consistent with the higher perceived cost of unstable deposit funding.

structure in the years after the shock as and a consequence grow slower in terms of size. The results suggest that the change in the funding structure slowed credit provision of high-exposed banks.

This paper informs the current policy debate on central bank digital currencies (CBDCs) and their potential effects on commercial banks. Similar to the contemporary proposals for CBDC, the deposit accounts at the Dutch central bank paid no interest and were intended as a means of payment, rather than a store of value. However, I show that when a central bank retail deposit account is designed as a means of payment, it can unintentionally become a store of value during crisis periods. Moreover, even when a crisis with corresponding deposit outflows at commercial banks is short-lived, these banks might reassess the costs of deposit funding and diversify away from deposits longer term. I provide suggestive evidence that such lower reliance on deposits can slow down the provision of credit to the real economy. A temporary flight to safety can thus have persistent consequences for credit provision.

Unlike current proposals, the Dutch central bank imposed no holding limits. Credible holding limits of CBDC can limit the outflow of deposits from commercial banks, but the counterfactual is not obvious: in 1914, many depositors moved into cash. Nowadays deposits might also find another store of value, such as money market funds, government bonds or stablecoins. Holding limits may therefore redirect a flight to safety rather than prevent it. Since a digital account is faster and cheaper to access than a physical account in 1914, the effects documented here are likely a lower bound.

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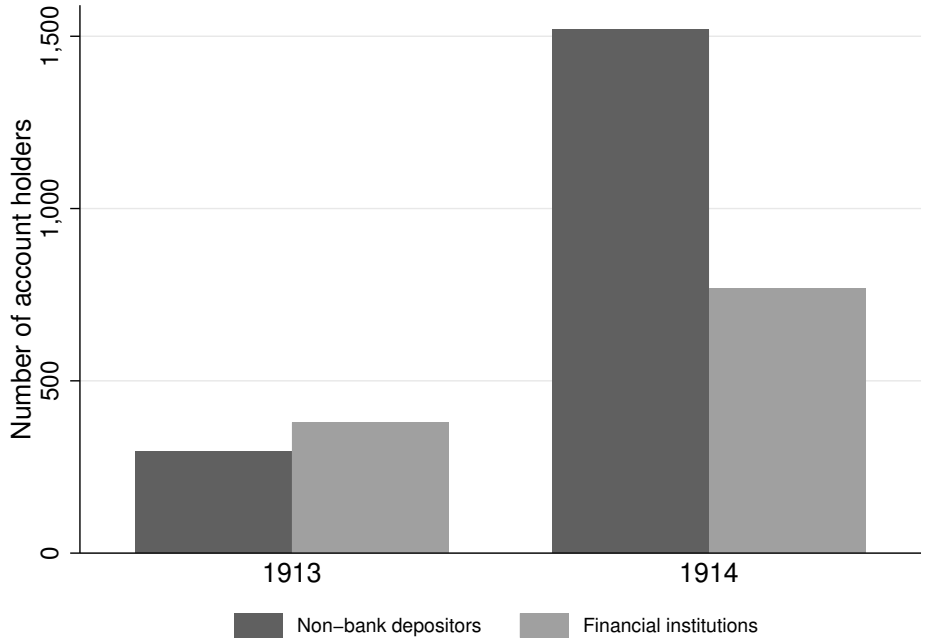
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Figures

Figure 1: Inflow of retail deposits at the central bank



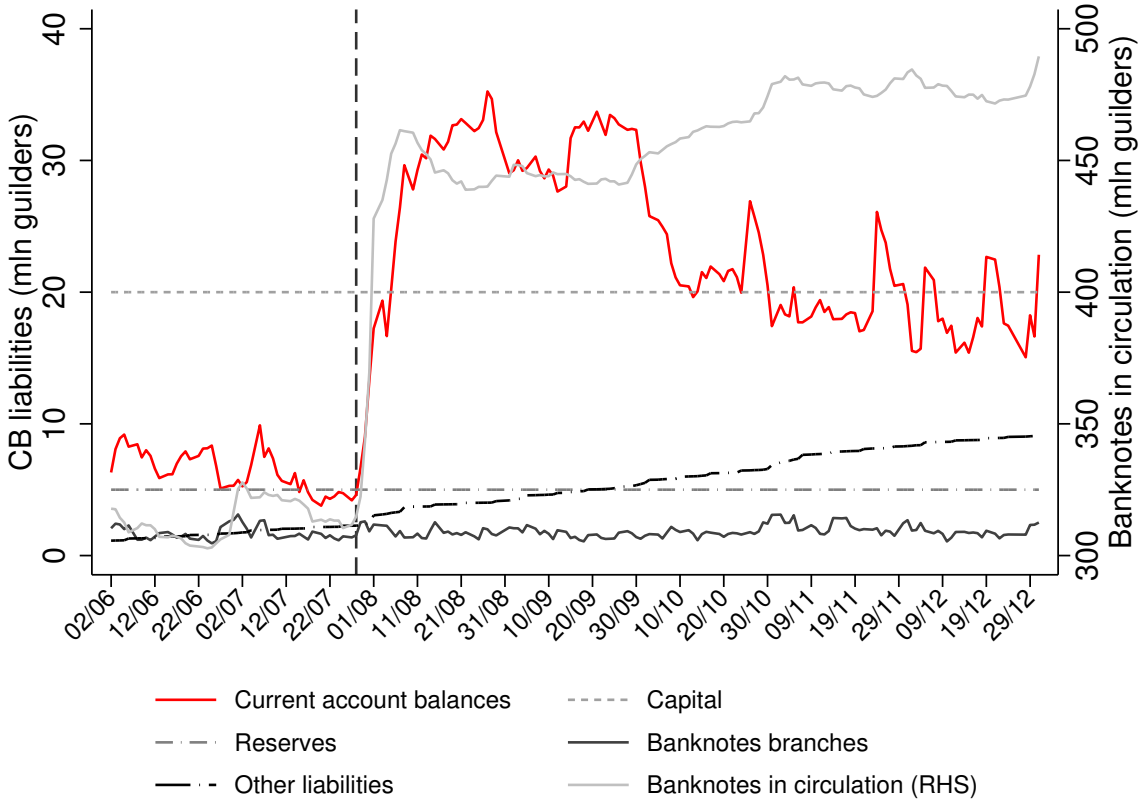
(a) Daily current account balances



(b) Composition of current account holders

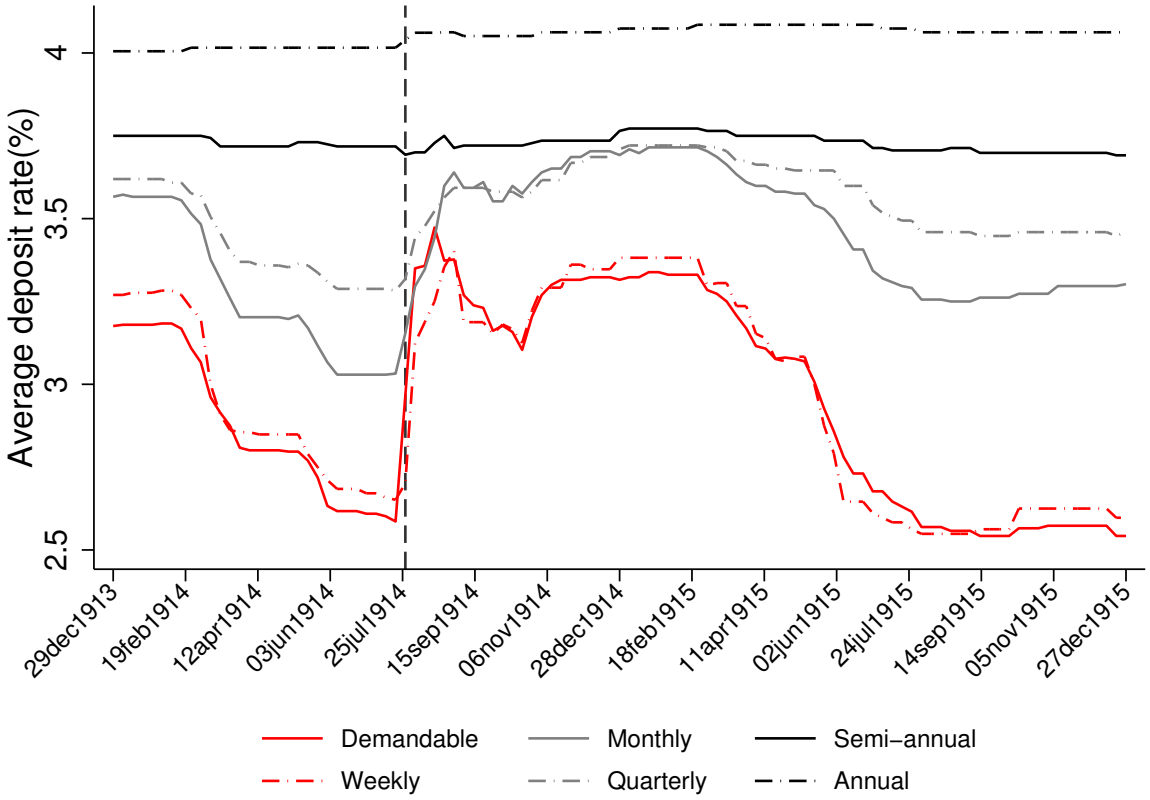
Note: Figure (a) shows the daily balance of current account holders (both banks and non-banks) at DNB from 20 July 1914 - 3 October 1914. The current account holders' balance is weighted by the size of the balance sheet before the outbreak of the war (*f*. 346 million on 20 July 1914). The vertical line reflects the outbreak of the war (28th of July 1914). Figure (b) shows the count and composition of the current account holders of DNB in 1913 and 1914. Names are manually matched to the Financial Address 1914 yearbook to identify financial institutions. Sources: DNB daily balance sheets, NA 2.25.68 - 100103; DNB Namelist of current account holders, NA 2.25.68 - 16431; Financieel Adresboek 1914.

Figure 2: DNB liabilities



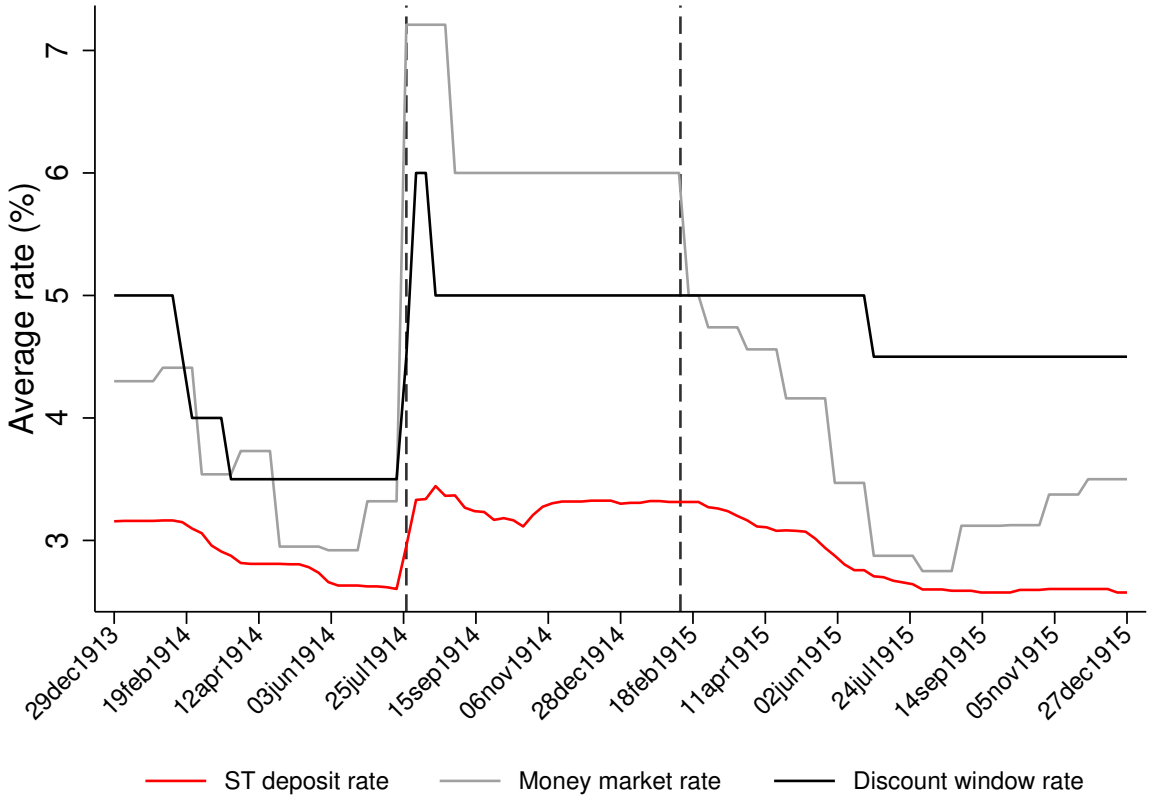
Note: This figure shows the daily breakdown of the liability side of DNB's balance sheet. All amounts are in million guilders. The vertical line reflects the outbreak of the war (28th of July 1914). Bank notes in circulation are displayed on the right-hand side axis for visibility. Source: DNB daily balance sheets, NA 2.25.68 - 100103.

Figure 3: Average commercial bank deposit rates



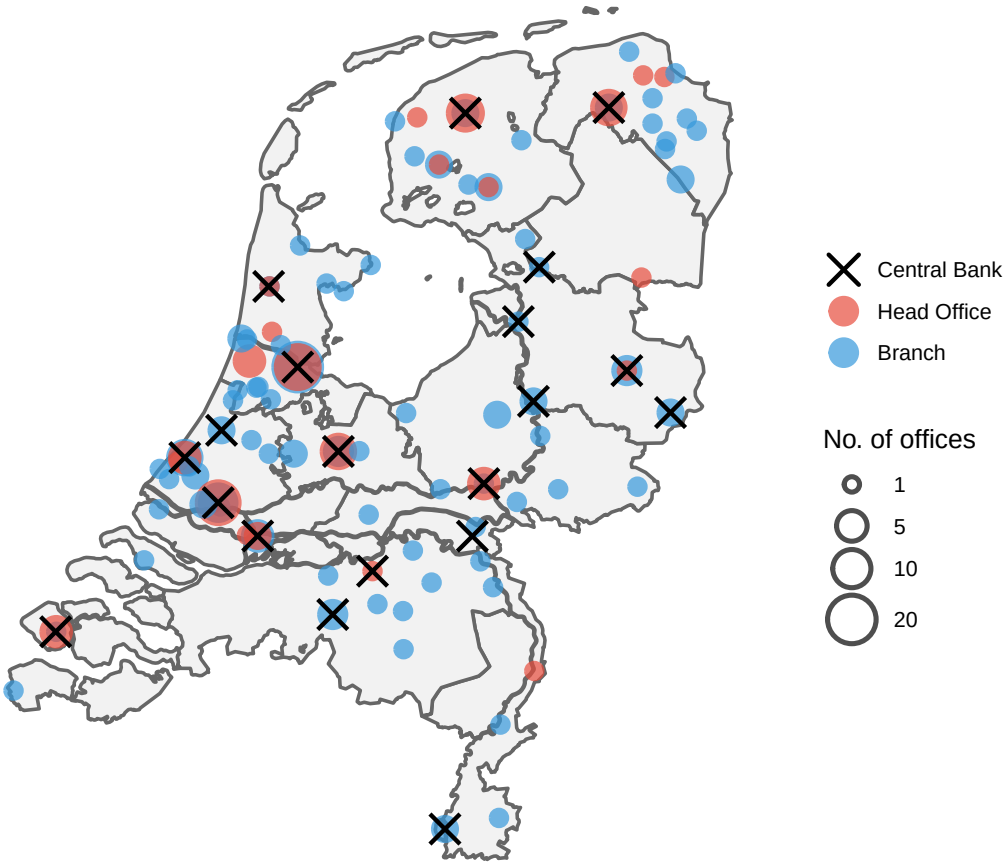
Note: This figure shows the weekly average deposit rate for the six most commonly used deposit tenors: demandable (0/1 day), weekly (8-14 days), monthly, quarterly(2-3 months), semi-annual and annual. All rates are simple averages across the sample of commercial banks from 1914-1915. The vertical line reflects the outbreak of the war (28th of July 1914). Source: Delpher Digital Newspaper Archive, author's own computations.

Figure 4: Average bank funding rates



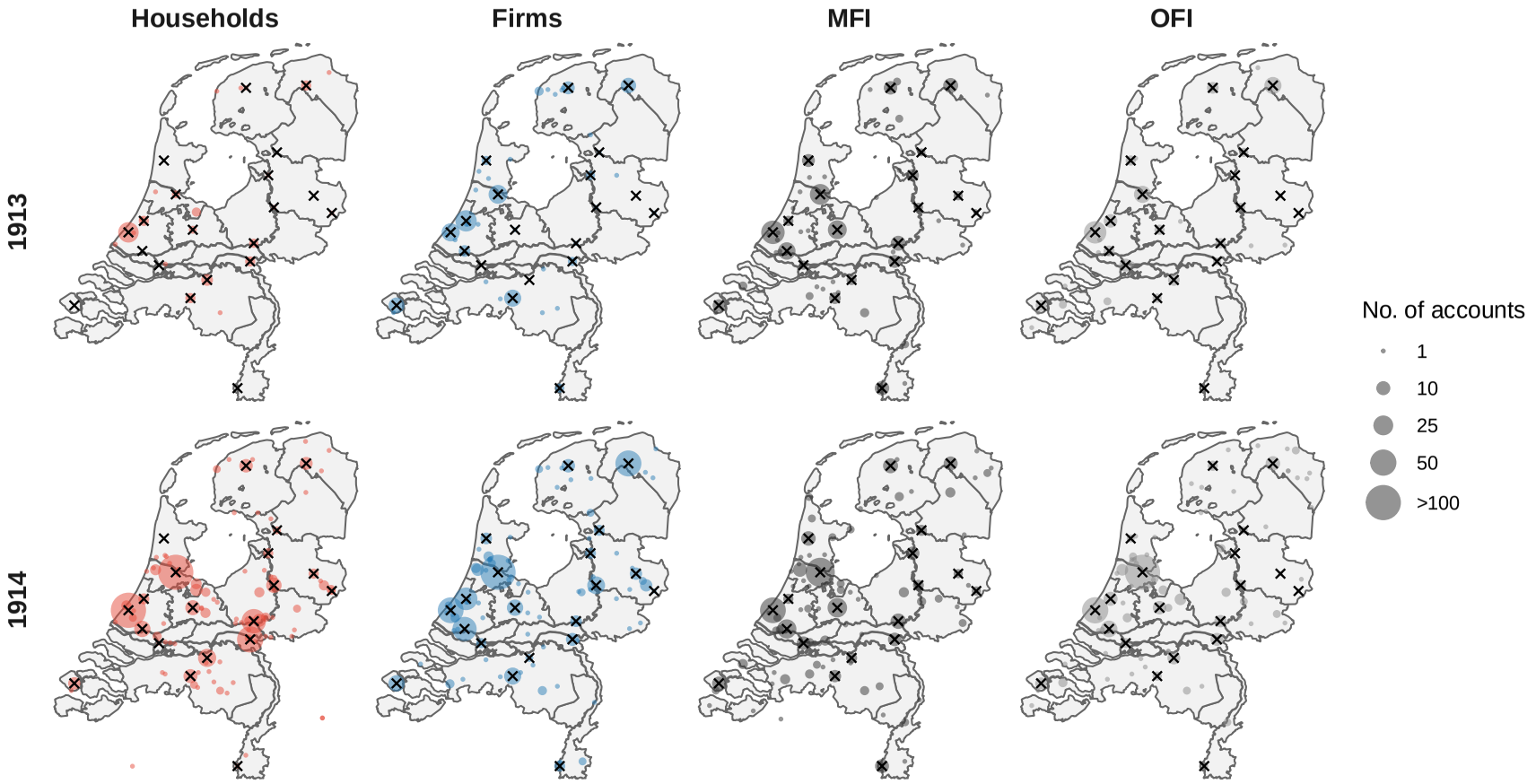
Note: This figure shows the weekly average short-term commercial bank deposit rate (red), the monthly repo market rate (grey) and the central bank discount window rate (black). The short-term commercial bank deposit rate is the simple average of the shortest rate offered by each commercial bank in the sample (only if a banks has demandable, daily or weekly). The vertical lines reflects the period that the stock market was closed (29th of July 1914 - 9 February 1915) and where the money market rate was artificially kept at 7/6%. Source: Delpher Digital Newspaper Archive, author's own computations; DNB archive NA 2.25.68, 7146; Tabellenboeken 1914-1915.

Figure 5: Geographical variation of commercial- and central bank branches



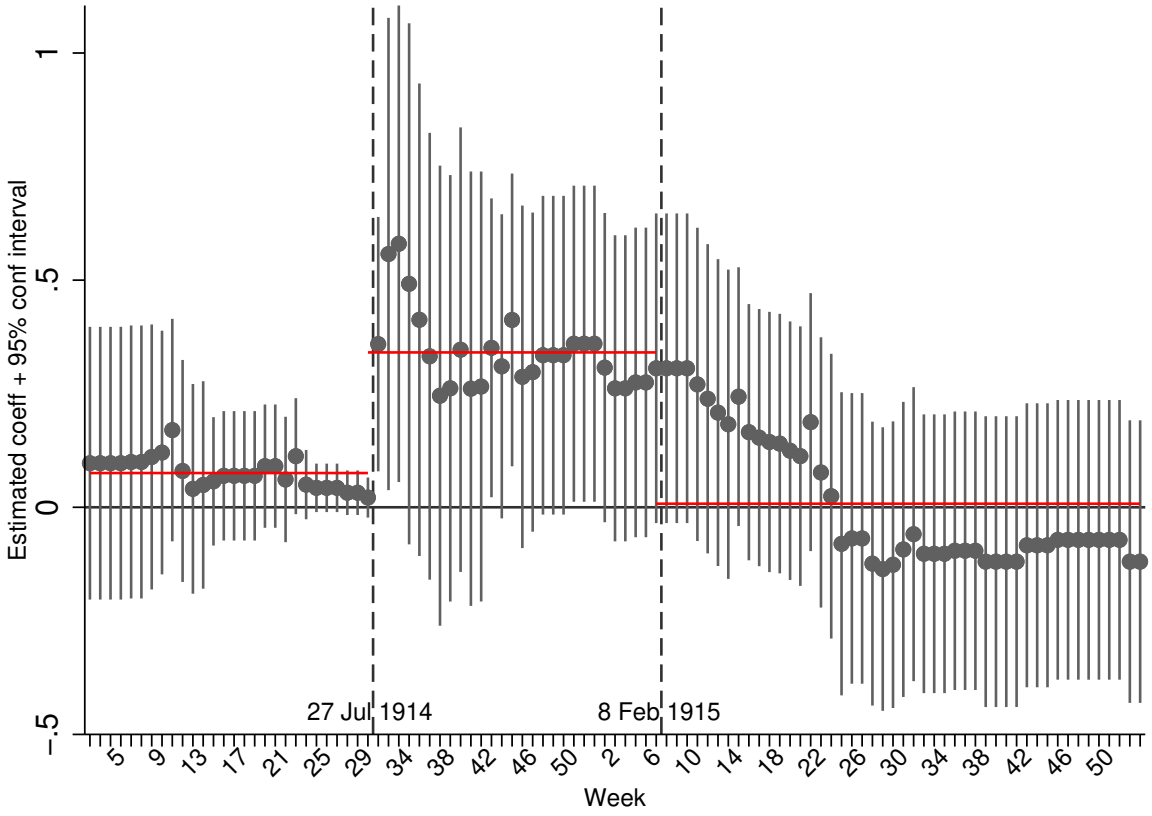
Note: This map shows the location of all commercial bank and central bank branches. The black crosses indicate the 20 central bank branches, the red dots are the locations of head offices of the commercial banks, and the blue dots are the location of commercial bank branches. Note that one dot can contain more than one bank located in the respective city. Includes all banks of the rate sample. Source: Financieel Adresboek 1914.

Figure 6: Geographical variation of DNB account holders, 1913 and 1914



Note: This map shows the location of all unique current account holders at the DNB offices, by sector (columns) and year (rows: 1913 above, 1914 below). Accounts are divided into households (red), non-financial companies (blue), banks (MFI, dark grey) and other financial intermediaries such as brokers, mortgage banks, insurance companies and savings banks (OFI, light grey). Bubble size is the number of unique accounts held in a town, on a common scale across all panels, capped at 100. The black crosses indicate the location of the 20 DNB offices. Source: DNB Namelist of current account holders, NA 2.25.68 - 16431.

Figure 7: Weekly coefficient plot 1914-1915

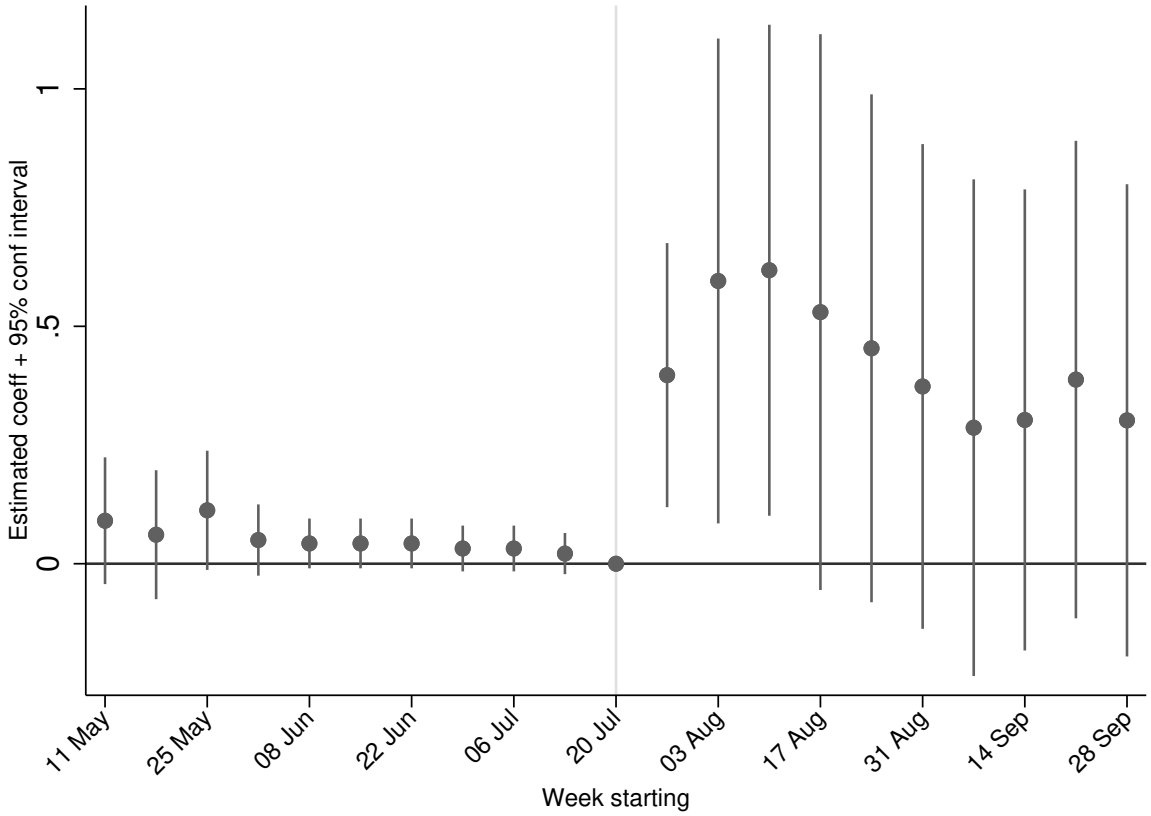


Note: This figure shows β coefficients of the following specification:

$$Y_{i,t} = \sum_{t=1914w1}^{t=1915w52} \beta_t \; CB \; Exposure_i \times D_{week=t} + \gamma Branches_{i,1913} \times Post_t + \kappa_i + \eta_t + \lambda_{g,t} + \epsilon_{i,t} \tag{9}$$

Where $Y_{i,t}$ is the short-term deposit rate of bank i in week t, $CB\ Exposure_i$ is the fraction of bank's i branches that are located in a city that has a central bank office, $D_{week=t}$ is a dummy equal to 1 for week t, where week 30 (the last week before the shock takes place) is left out to avoid the dummy variable trap. The period covered is all weeks in 1914-1915. The control variable Branches_i is the number of offices bank i operated before the shock. Both CB exposure_i and Branches_i are time-invariant and are therefore absorbed by the bank fixed effects α_i; only their interactions with Post_t are identified. Bank i, week t and region $\times$ week g, t fixed effects are included, and standard errors are clustered on the bank level. The short-term commercial bank deposit rate is the simple average of the shortest rate offered by each commercial bank in the sample (only if a banks has demandable, daily or weekly). The vertical dashed lines reflect the outbreak of the war (28th of July 1914) and the reopening of the Amsterdam Stock Exchange (9th of February 1915). Source: Delpher Digital Newspaper Archive, author's own computations.

Figure 8: Parallel trends: weekly coefficients are not significantly different before the shock

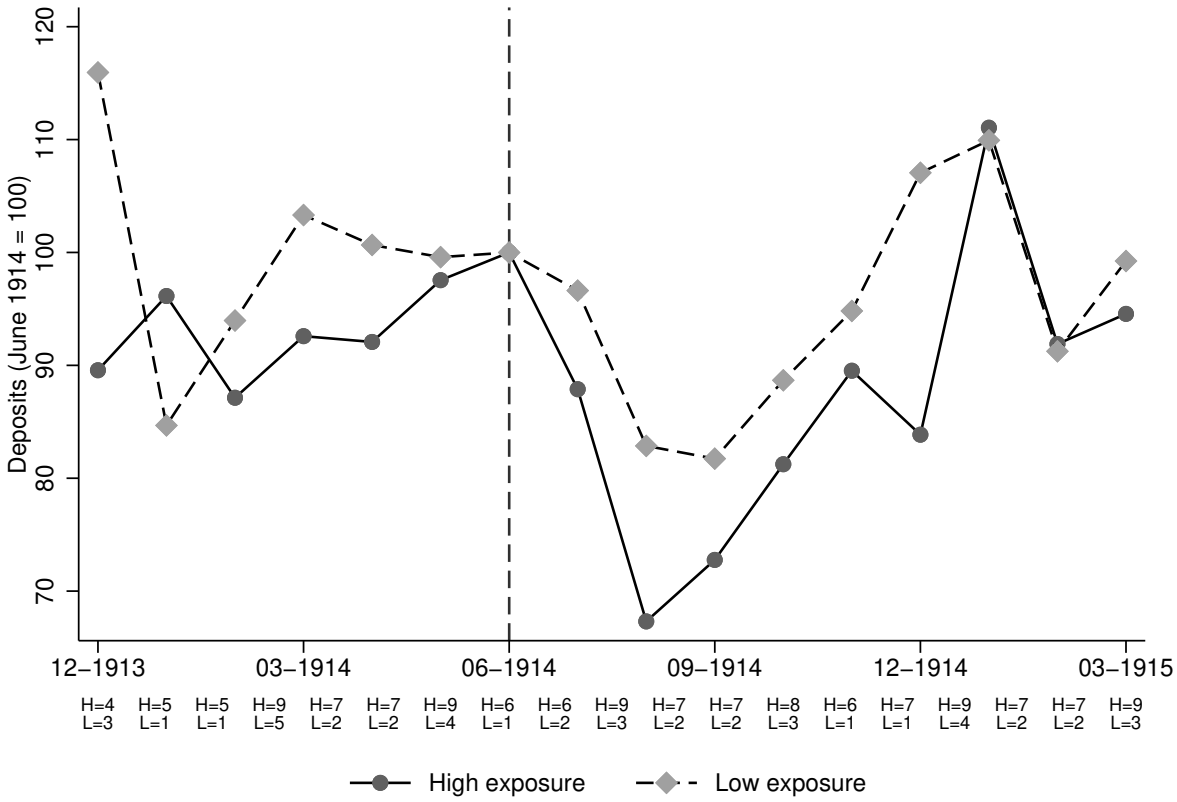


Note: This figure shows β coefficients of the following specification:

$$Y_{i,t} = \sum_{t=1914w20}^{t=1914w40} \beta_t \; CB \; Exposure_i \times D_{week=t} + \gamma Branches_{i,1913} \times Post_t + \kappa_i + \eta_t + \epsilon_{i,t} \tag{11}$$

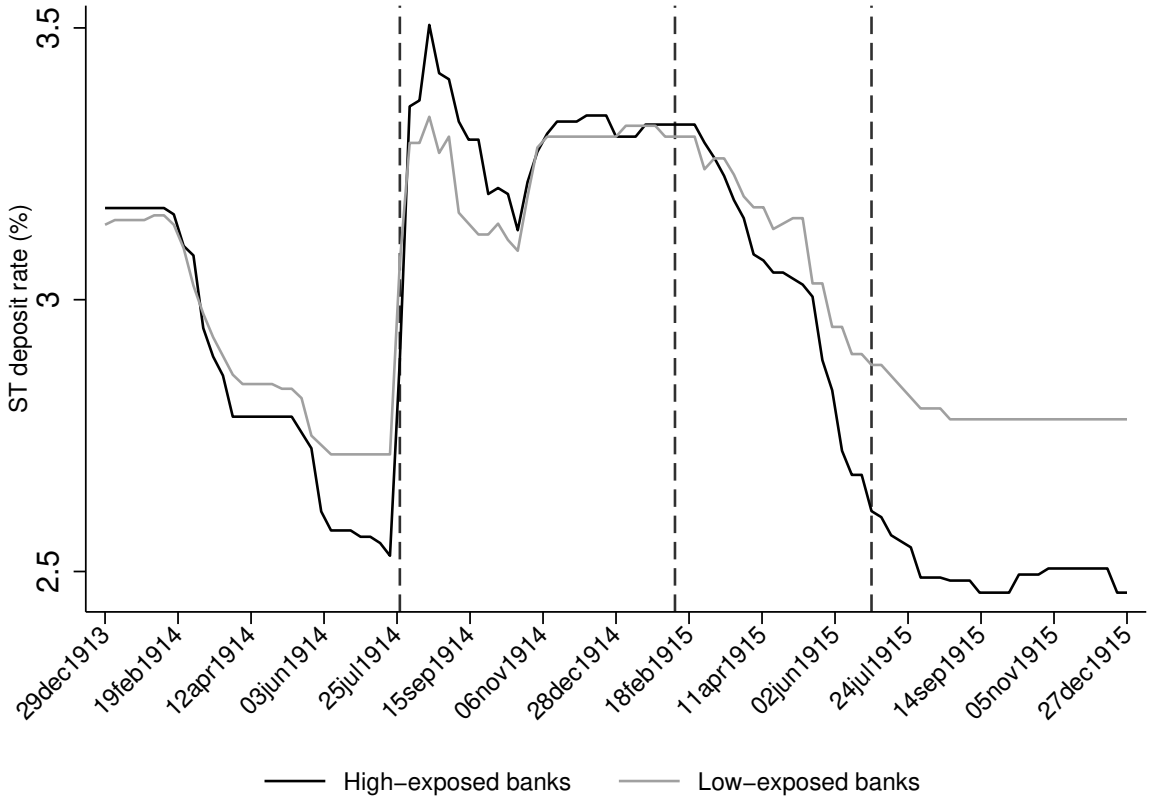
Where $Y_{i,t}$ is the short-term deposit rate of bank i in week t, $CB\ Exposure_i$ is the fraction of bank's i branches that are located in a city that has a central bank office, $D_{week=t}$ is a dummy equal to 1 for week t, where week 30 (the last week before the shock takes place) is left out to avoid the dummy variable trap. The period covered is from mid-May - September 1914. Bank and week fixed effects are included. The control variable Branches_i is the number of offices bank i operated before the shock. Both CB exposure_i and Branches_i are time-invariant and are therefore absorbed by the bank fixed effects α_i; only their interactions with Post_t are identified. Standard errors are clustered on the bank level. The short-term commercial bank deposit rate is the simple average of the shortest rate offered by each commercial bank in the sample (only if a banks has demandable, daily or weekly). The vertical line reflects the outbreak of the war (28th of July 1914). Source: Delpher Digital Newspaper Archive, author's own computations.

Figure 9: Deposits return to banks within the year



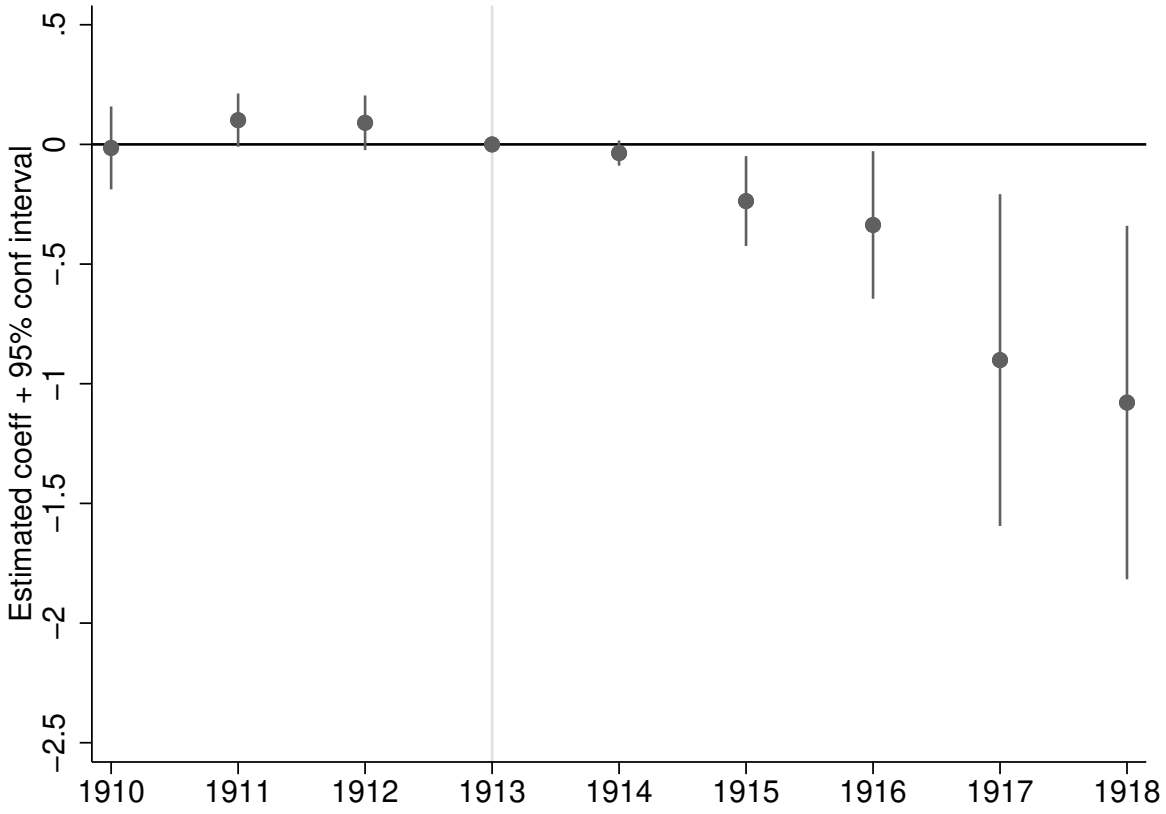
This figure shows the unweighted average of bank-level deposit indices for high- (full) and low-exposed banks. Each bank's index is set to 100 for the last pre-war statement. The vertical dashed line marks the last observation before the shock. Deposits are from monthly and quarterly statements of banks who published those in newspapers. Each point averages over the banks that published a statement in that month. Only few banks did so; the number per data point is provided below the horizontal axis. One bank (Noord-Hollandsch Landbouwcrediet) published the exact same quarterly financial statement for June and September 1914 and is therefore excluded. Source: Delpher Digital Newspaper Archive, author's own computations.

Figure 10: More exposed banks especially lower their rates after the reopening of the stock exchange



Note: This figure shows the unweighted average short-term deposit rate for high and low-exposed commercial banks at weekly frequency. High exposed commercial banks are above median (= fully) exposed to the central bank, meaning all their branches are located in a city with a DNB office. Low-exposed commercial banks have one or more branches located outside a central bank city. The short-term commercial bank deposit rate is the simple average of the shortest rate offered by each commercial bank in the sample (only if a banks has demandable, daily or weekly). The vertical lines reflect the outbreak of the war (28th of July 1914), the reopening of the stock market (9th of February 1915) and the settlement of frozen repos (1st July 1915). Source: Delpher Digital Newspaper Archive, author's own computations.

Figure 11: Yearly coefficients of banks more exposed to the central bank



Note: This figure shows β coefficients of the following specification:

$$Y_{i,t} = \sum_{t=1910}^{t=1918} \beta_t \; CB \; Exposure_i \times D_{year=t} + \kappa_i + \eta_t + \epsilon_{i,t} \tag{13}$$

Where $Y_{i,t}$ is the deposit ratio over pre-war (1913) total assets of bank i in year t, $CB\ Exposure_i$ is the fraction of bank's i branches that are located in a city that has a central bank office, $D_{year=t}$ is a dummy equal to 1 for year t, where 1913 (the last year before the shock takes place) is left out to avoid the dummy variable trap. The period covered is 1910-1918. Bank, year and CB ressort $\times$ year fixed effects are included, and standard errors are clustered on the bank level. Source: DNB archive, Dutch commercial balance sheets, author's own computations.

Tables

Table 1: Summary statistics: deposit rate sample

	Mean	SD	P1	P50	P99	N	Banks
Short-term rate	2.95	0.80	1.00	3.00	4.00	7414	76
Demandable rate	2.95	0.81	1.00	3.00	4.00	6795	68
Weekly rate	2.96	0.80	0.75	3.00	4.50	3840	39
Monthly rate	3.42	0.73	1.25	3.50	5.00	4639	50
Quarterly rate	3.53	0.55	2.25	3.50	4.50	4609	48
Yearly rate	4.05	0.23	3.50	4.00	5.00	4714	48
Inverted term structure	0.08	0.26	0.00	0.00	1.00	7444	77
CB exposure	0.74	0.38	0.00	1.00	1.00	7579	79
High CB exposure	0.62	0.49	0.00	1.00	1.00	7579	79
War exposure	0.19	0.36	0.00	0.00	1.00	7579	79
# branches	2.73	3.03	1.00	1.00	16.00	7579	79
Log(population)	12.12	1.54	8.57	12.51	15.12	7579	79
Unemployment rate	61.01	1.66	55.28	60.92	66.77	7579	79
# Banks	29.06	20.73	1.00	23.00	62.00	7579	79

Note: This table reports summary statistics for the weekly deposit rate panel of commercial banks, 1913–1915. N is the number of bank-week observations and Banks the number of banks with a non-missing value. Short-term rate is the shortest rate available for each bank that is either demandable, daily or weekly. Demandable rate is the deposit rate for deposits with a tenor of 0 or 1 day. The weekly rate includes deposit tenors of 8, 10 and 14 days. Monthly, Quarterly and Yearly rates refer to one month, 2-3 months and 12 months deposit tenors. Inverted term structure is a dummy that equals 1 if a bank remunerates a shorter tenor at a higher rate than a longer tenor, and 0 otherwise. CB exposure is the continuous fraction of commercial bank branches that is located in a city that has a central bank office. High CB exposure is a dummy that equals 1 if a bank is fully exposed to the central bank, i.e. all its branches are located in CB cities. War exposure is the fraction of commercial bank branches that are located in a province bordering countries at war. # branches is the number of offices a bank has pre-shock. Log(population) is the natural logarithm of the total population a bank serves, i.e. the sum of the inhabitants of every branch location. Unemployment rate is the population-weighted average percentage unemployed across a bank's branches. # Banks is the population-weighted average number of banks in a bank's branch locations.

Table 2: Bank balancing table: deposit rates sample

	Low CB exposure		High CB exposure		Pairwise t-test
	N	Mean/(SD)	N	Mean/(SD)	N	t-statistic
Panel A: exposure and pre-war deposit pricing						
CB exposure	29	0.300	43	1.000	72	-16.218***
		(0.284)		(0.000)		
War exposure	29	0.265	43	0.128	72	1.596
		(0.400)		(0.326)		
Branch network size	29	4.241	43	1.791	72	3.624***
		(3.280)		(2.455)		
Short-term rate	29	2.911	43	2.858	72	0.428
		(0.500)		(0.517)		
Demandable rate	23	2.942	41	2.842	64	0.756
		(0.493)		(0.519)		
Weekly rate	15	3.046	23	2.877	38	1.250
		(0.421)		(0.400)		
Monthly rate	19	3.333	26	3.228	45	0.949
		(0.344)		(0.384)		
Quarterly rate	18	3.499	26	3.382	44	1.101
		(0.350)		(0.343)		
Yearly rate	20	4.025	26	4.004	46	0.553
		(0.112)		(0.134)		
Inverted term structure	28	0.048	43	0.050	71	-0.042
		(0.195)		(0.195)		
Panel B: 1913 balance sheets						
Log(total assets)	18	15.719	24	15.865	42	-0.309
		(1.624)		(1.427)		
Deposits/total assets	18	0.306	24	0.359	42	-0.961
		(0.131)		(0.203)		
Secured funding/total assets	18	0.144	24	0.158	42	-0.269
		(0.153)		(0.183)		
Unsecured funding/total assets	18	0.259	24	0.152	42	1.907*
		(0.190)		(0.173)		
Total equity/total assets	18	0.256	23	0.269	41	-0.451
		(0.105)		(0.092)		
Repo lending/total assets	18	0.159	24	0.123	42	0.670
		(0.199)		(0.149)		
Liquid assets/total assets	18	0.026	24	0.032	42	-0.773
		(0.014)		(0.029)		

Note: The balancing table compares low-exposed and high-exposed banks before the outbreak of the war. N is the number of banks. Panel A uses the banks in the deposit rate sample that quote a short-term rate before 27 July 1914, i.e. the banks that identify the deposit rate regressions: each bank's weekly observations before that date are first averaged, so every bank counts once and standard deviations are across banks. High CB exposure is a dummy that equals 1 if a bank is fully exposed to the central bank, i.e. all its branches are located in CB cities. CB exposure is the continuous fraction of commercial bank branches that is located in a city that has a central bank office. War exposure is the fraction of commercial bank branches that are located in a province bordering countries at war. Branch network size is the number of offices a bank has pre-shock. Short-term rate is the shortest rate available for each bank that is either demandable, daily or weekly. Demandable rate is the deposit rate for deposits with a tenor of 0 or 1 day. The weekly rate includes deposit tenors of 8, 10 and 14 days. Monthly, Quarterly and Yearly rates refer to one month, 2-3 months and 12 months deposit tenors. Inverted term structure is the share of pre-war weeks in which a bank remunerated a shorter tenor at a higher rate than a longer tenor. Panel B uses the 1913 balance sheets of the same banks, where available; all balance sheet variables are weighted by total assets. The last column reports the t-statistic of a two-sided test of equal means.

Table 3: Commercial bank short-term deposit rates and central bank exposure

	(1)	(2)	(3)	(4)	(5)	(6)
	Demandable deposit rate					
CB exposure * Post	0.458**	0.372*	0.423**	0.618***	0.538***	0.551***
	(0.204)	(0.197)	(0.211)	(0.191)	(0.185)	(0.197)
# branches * Post				0.060***	0.057***	0.048***
				(0.022)	(0.016)	(0.016)
Observations	640	640	640	640	640	640
R-squared	0.610	0.636	0.669	0.621	0.646	0.676
Within R-sq	0.025	0.017	0.022	0.052	0.043	0.041
Banks	76	76	76	76	76	76
Period	Jul-Aug 1914					
Bank FE	Y	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y
Region-Month FE	N	Y	N	N	Y	N
Ressort-Month FE	N	N	Y	N	N	Y

Note: This table presents estimates for the demandable deposit rates of commercial banks in the month before (July) and the month after (August) the shock. The dependent variable is the shortest deposit rate available for each bank (max weekly rate). Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All columns include bank and week fixed effects. Columns (2) and (5) also include region x month fixed effects, where the regions are north, middle, south and 'Randstad' (North- and South-Holland). Columns (3) and (6) include central bank office region (see Figure B.1) x month fixed effects. Columns (4)-(6) control for the number of branches, the number of offices a bank has pre-shock, interacted with Post. Table C.1 in the appendix reports the same estimates over 1914 and 1914-1915. Clustered standard errors on the bank level are in parentheses. Wild cluster bootstrap p-values (Rademacher weights, 9,999 replications) for CB exposure x Post range from 0.006 to 0.079 across columns. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 4: Central bank exposure, regional controls and comparable banks, demandable deposit pricing

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Demandable rate								
CB exposure * Post	0.372*	0.538***	0.340*	0.350*	0.350*	0.361	0.454***	0.478***	0.526***
	(0.197)	(0.185)	(0.184)	(0.191)	(0.189)	(0.246)	(0.148)	(0.169)	(0.150)
# branches * Post		0.057***	0.030	0.033*	0.033	0.066***			0.049**
		(0.016)	(0.019)	(0.019)	(0.023)	(0.021)			(0.021)
Log(population) * Post			0.111**	0.105*	0.101				
			(0.055)	(0.054)	(0.102)				
Unemployment rate * Post				-0.059	-0.058	-0.068			0.006
				(0.043)	(0.039)	(0.041)			(0.041)
# Banks * Post					0.000	0.004			0.006*
					(0.007)	(0.006)			(0.004)
Observations	640	640	640	640	640	349	640	583	583
R-squared	0.636	0.646	0.650	0.652	0.652	0.688	0.642	0.643	0.656
Within R-sq	0.017	0.043	0.054	0.060	0.060	0.088	0.020	0.020	0.056
Banks	76	76	76	76	76	41	76	67	67
Period	Jul-Aug 1914								
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Region-Month FE	Y	Y	Y	Y	Y	Y	N	N	N
Sector-Month FE	N	N	N	N	N	N	Y	Y	Y
Common support	N	N	N	N	N	Y	N	N	N
Excl. agriculture	N	N	N	N	N	N	N	Y	Y

Note: This table presents estimates for the demandable deposit rates of commercial banks with regional controls and restricted to comparable banks, in the month before (July) and the month after (August) the shock. Table **??** in the appendix reports the same estimates for all of 1914. The dependent variable is the shortest deposit rate available for each bank (max weekly rate). Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. Columns (1)-(5) add the controls one at a time. # branches is the number of offices a bank has pre-shock. This excludes branches outside of the Netherlands (3 banks have an office in London or Batavia). Log(population) is the natural logarithm of the total population that a bank serves, i.e. the sum of all city inhabitants of every branch location. Unemployment rate is the population-weighted average percentage unemployed across a banks' branches. # Banks is the average number of banks (population-weighted) in its branch locations. All controls are interacted with Post. Column (6) includes only banks in the common support of log average population across the branches, i.e. between the larger of the minima and the smaller of the maxima of the high- and low-exposed groups. Columns (7)-(9) include dominant sector x month fixed effects instead of region x month fixed effects; columns (8) and (9) exclude the sector that only has low-exposed banks (agriculture). Columns (1)-(6) include region x month fixed effects, where the regions are north, middle, south and 'Randstad' (North- and South-Holland). All columns include bank and week fixed effects. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 5: Commercial bank deposit rates by tenor and central bank exposure

	(1) Demandable	(2) Weekly	(3) Monthly	(4) Quarterly	(5) Annual
CB exposure * Post	0.538***	0.637***	0.183	0.183	0.032
	(0.185)	(0.197)	(0.245)	(0.149)	(0.033)
# branches * Post	0.057***	0.054	0.038	0.017	-0.001
	(0.016)	(0.037)	(0.025)	(0.017)	(0.002)
Observations	640	330	403	400	409
R-squared	0.646	0.690	0.721	0.735	0.878
Within R-sq	0.043	0.041	0.019	0.017	0.007
Banks	76	39	50	48	48
Period			Jul-Aug 1914		
Bank FE	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y
Region-Month FE	Y	Y	Y	Y	Y

Note: This table presents estimates for the deposit rates of commercial banks in July and August 1914 by different deposit tenors. Column (1) includes only the shortest deposit rate available (max tenor is weekly). Column (2), the weekly rate, includes deposit tenors of 8, 10 and 14 days. Columns (3)-(5) show results for monthly, quarterly and annual deposit tenors, respectively. All columns control for the number of branches, the number of offices a bank has pre-shock, interacted with Post. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All columns include bank, week and region x month fixed effects, where the regions are north, middle, south and 'Randstad' (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 6: Central bank exposure and inverted term structure of deposit pricing

	(1)	(2)	(3)	(4)	(5)	(6)
	Inverted term structure					
CB exposure * Post	0.192***	0.201***	0.169**	0.242***	0.261***	0.225***
	(0.067)	(0.071)	(0.069)	(0.073)	(0.079)	(0.074)
# branches * Post				0.019	0.021*	0.021*
				(0.013)	(0.012)	(0.012)
Observations	643	643	643	643	643	643
R-squared	0.603	0.612	0.625	0.608	0.618	0.631
Within R-sq	0.021	0.022	0.015	0.034	0.038	0.031
Banks	77	77	77	77	77	77
Period	Jul-Aug 1914					
Bank FE	Y	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y
Region-Month FE	N	Y	N	N	Y	N
Ressort-Month FE	N	N	Y	N	N	Y

Note: This table presents estimates for inverted term structure of deposit rates by commercial banks in the month before (July) and the month after (August) the shock. Inverted term structure is a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable rate is higher than weekly rate and/or weekly rate is higher than monthly, or demandable rate is higher than monthly rate), and 0 otherwise; banks that quote only one of these tenors are coded 0. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All columns include bank and week fixed effects. Columns (2) and (5) also include region x month fixed effects, where the regions are north, middle, south and 'Randstad' (North- and South-Holland). Columns (3) and (6) include central bank office region (see Figure B.1) x month fixed effects. Columns (4)-(6) control for the number of branches, the number of offices a bank has pre-shock, interacted with Post. Table C.1 in the appendix reports the same estimates over 1914 and 1914-1915. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 7: Central bank exposure and deposit rates after the closure of the exchange

	(1)	(2)	(3)	(4)	(5)
	Demandable deposit rate				
CB exposure * Exchange closed	0.339**	0.397***	0.398***	0.367***	0.369***
	(0.131)	(0.126)	(0.127)	(0.130)	(0.132)
CB exposure * Exchange reopened	0.120	0.355*	0.360**	0.291	0.301
	(0.142)	(0.178)	(0.179)	(0.177)	(0.182)
CB exposure * Repos unfrozen	-0.269	-0.178	-0.169	-0.318	-0.300
	(0.179)	(0.223)	(0.211)	(0.198)	(0.192)
Amsterdam office * Exchange closed			-0.118		-0.147
			(0.240)		(0.236)
Amsterdam office * Exchange reopened			-0.497**		-0.498**
			(0.213)		(0.237)
Amsterdam office * Repos unfrozen			-0.874***		-0.863***
			(0.239)		(0.275)
Large bank * Exchange closed				0.046	0.048
				(0.101)	(0.102)
Large bank * Exchange reopened				-0.262**	-0.256**
				(0.105)	(0.111)
Large bank * Repos unfrozen				-0.587***	-0.577***
				(0.131)	(0.139)
p-value: unfrozen = reopened	0.013	0.001	0.001	0.000	0.000
Observations	6,897	6,897	6,897	6,656	6,656
R-squared	0.785	0.795	0.804	0.813	0.821
Banks	67	67	67	64	64
Period			1914-1915		
Branch controls	Y	Y	Y	Y	Y
Controls	N	Y	Y	Y	Y
Bank FE	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y
Region-Month FE	Y	Y	Y	Y	Y

Note: This table presents estimates of the demandable deposit rate (in percentage points) for 1914-1915, for banks observed both before and after 27 July 1914. Exchange closed is a dummy that equals 1 from 27 July 1914 to 1 February 1915, Exchange reopened from 8 February to 21 June 1915, and Repos unfrozen from the week of 28 June 1915, when frozen repos were released (1 July) and the DNB lowered its discount rate (28 June). Each coefficient is measured against the pre-war weeks (5 January-20 July 1914). CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. Branch controls are the size of the bank's branch network interacted with each phase; Controls are the log population of the bank's office locations, the unemployment rate and the number of banks, each interacted with each phase. Amsterdam office is a dummy that equals 1 if the bank has an office in Amsterdam; as Amsterdam is a CB city, the CB exposure coefficients in columns (3) and (5) capture exposure beyond an Amsterdam office. Large bank is a dummy that equals 1 if the bank's 1913 total assets are above the median, hand-coded where no balance sheet is available; columns (4) and (5) use the banks with a size coding. p-value: unfrozen = reopened tests whether the CB exposure coefficient changed between the reopened and the unfrozen phase. Wild cluster bootstrap p-values (Rademacher weights, 9999 replications), range across columns: CB exposure x Exchange closed: 0.011–0.025; CB exposure x Exchange reopened: 0.099–0.434; CB exposure x Repos unfrozen: 0.131–0.453; unfrozen = reopened: 0.001–0.030. All columns include bank, week and region x month fixed effects. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 8: Summary statistics: balance sheet sample

	Mean	SD	P1	P50	P99	N	Banks
Total assets	1.31	0.87	0.49	1.00	5.17	522	70
Deposits	0.51	0.58	0.00	0.38	2.39	523	70
Secured funding	0.12	0.19	0.00	0.04	0.88	523	70
Unsecured funding	0.25	0.36	0.00	0.11	1.72	523	70
Equity	0.32	0.22	0.03	0.27	1.23	510	69
Other liabilities	0.12	0.29	0.00	0.02	1.67	492	68
Total lending	0.92	0.66	0.06	0.78	3.60	523	70
Liquid lending	0.16	0.27	0.00	0.06	1.41	523	70
Illiquid lending	0.63	0.55	0.00	0.53	2.57	523	70
Repo lending	0.13	0.19	0.00	0.04	0.85	523	70
Liquid assets	0.06	0.08	0.00	0.03	0.45	523	70
CB exposure	0.71	0.40	0.00	1.00	1.00	531	72
High CB exposure	0.61	0.49	0.00	1.00	1.00	531	72
# branches	2.96	3.22	1.00	1.00	16.00	531	72
Log(population), yearly	12.37	1.52	8.84	12.62	15.26	531	72
Large bank	0.53	0.50	0.00	1.00	1.00	518	68

Note: This table reports summary statistics for the annual balance sheet panel used in the funding and lending regressions, 1910–1918. N is the number of bank-year observations and Banks the number of banks with a non-missing value. All balance sheet variables are divided by the bank's 1913 (pre-war) total assets, as in the regressions. Total deposits include current account balances; Secured funding is short-term collateralized borrowing (repo and bills of exchange); Unsecured funding is interbank borrowing, creditors and long-term borrowing; Equity is capital and retained earnings; Other liabilities is the residual on the liability side. Liquid lending consists of bills of exchange, Illiquid lending of credit lines and long-term loans, and Repo lending of collateralized loans on securities; Liquid assets include cash and cash equivalents and Dutch government bonds. CB exposure is the continuous fraction of commercial bank branches that is located in a city that has a central bank office. High CB exposure is a dummy that equals 1 if all of a bank's branches are located in CB cities. # branches is the number of offices a bank has pre-shock. Log(population), yearly, is the natural logarithm of the total population a bank serves in each year. Large bank is a dummy that equals 1 for banks above the median 1913 size.

Table 9: Bank balancing table: balance sheet sample

	Low CB exposure		High CB exposure			Pairwise comparison
	N	Mean/(SD)	N	Mean/(SD)	N	Mean difference
CB exposure	28	0.258	40	1.000	68	-0.742***
		(0.271)		(0.000)		
War exposure	28	0.162	40	0.088	68	0.074
		(0.338)		(0.271)		
Log(total assets)	28	15.048	40	15.323	68	-0.275
		(1.622)		(1.463)		
Deposits/total assets	28	0.327	40	0.349	68	-0.022
		(0.204)		(0.236)		
Secured funding/total assets	28	0.144	40	0.140	68	0.004
		(0.166)		(0.163)		
Unsecured funding/total assets	28	0.242	40	0.185	68	0.057
		(0.232)		(0.245)		
Total equity/total assets	28	0.277	38	0.262	66	0.016
		(0.128)		(0.126)		
Liquid lending/total assets	28	0.103	40	0.106	68	-0.002
		(0.143)		(0.138)		
Illiquid lending/total assets	28	0.592	40	0.482	68	0.109*
		(0.231)		(0.277)		
Repo/total assets	28	0.129	40	0.121	68	0.008
		(0.179)		(0.149)		
Liquid assets/total assets	28	0.029	40	0.027	68	0.002
		(0.020)		(0.026)		

Note: The balancing table compares the pre-war (before 1914) annual variable means of low-exposed and high-exposed banks. The balancing table covers banks observed in the pre-period. High CB exposure is a dummy that equals 1 if a bank is fully exposed to the central bank, i.e. all its branches are located in CB cities. CB exposure is the continuous fraction of commercial bank branches that is located in a city that has a central bank office. Log(total assets) is the natural log of a bank's total assets in 1913. Total deposits is the total deposits and current account outstanding balances, Secured funding is all short-term collateralized borrowing (repo and bills of exchange), Unsecured funding is the sum of interbank borrowing, creditors and long-term borrowing, Equity is the sum of capital and retained earnings. Liquid lending consists of bills of exchange, while illiquid lending contains credit lines and long-term loans. Liquid assets include cash and cash equivalents and Dutch government bonds. All balance sheet variables are 1913 values weighted by total assets. War exposure is the fraction of commercial bank branches that are located in a province bordering countries at war. The last column reports the difference in means between the two groups.

Table 10: Central bank exposure and commercial banks' funding mix

	(1) Total Assets	(2) Deposits	(3) Secured funding	(4) Unsecured funding	(5) Equity	(6) Other liabilities
CB exposure * Post	-0.414*	-0.426***	0.001	0.027	-0.125**	0.140*
	(0.224)	(0.147)	(0.048)	(0.077)	(0.059)	(0.071)
# branches * Post	0.014	-0.002	-0.001	0.008	0.004	0.011
	(0.031)	(0.011)	(0.006)	(0.012)	(0.006)	(0.012)
Log(population), yearly	6.570***	7.261***	0.170	-0.900**	0.232	-0.274
	(1.650)	(1.354)	(0.149)	(0.385)	(0.295)	(0.367)
Pre-shock mean	0.943	0.306	0.122	0.196	0.272	0.054
Post-shock mean	1.617	0.681	0.111	0.305	0.368	0.170
Observations	522	523	523	523	510	492
R-squared	0.670	0.786	0.697	0.738	0.657	0.535
Banks	70	70	70	70	69	68
Period				1910-1918		
Bank FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Region-Year FE	Y	Y	Y	Y	Y	Y

Note: This table presents annual bank balance sheet items for 1910-1918. Six banks in the balance sheet sample do not report deposits in more than one year and therefore drop out of the deposits column. Post is a dummy that equals 1 if the year is 1914 or later. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All dependent variables are weighted by their 1913 (pre-war) total assets. Total Asset is thus the total assets of a bank i in year t over total assets of 1913. Total deposits is the total deposits and current account outstanding balances, Secured funding is all short-term collateralized borrowing (repo and bills of exchange), Unsecured funding is the sum of interbank borrowing, creditors and long-term borrowing, Equity is the sum of capital and retained earnings, and Other Liabilities reflects a residual term on the banks' balance sheet. All columns control for the log of the yearly population of the bank's pre-war office locations and for the size of its branch network interacted with Post. All columns include bank, year and region x year fixed effects, where the regions are north, middle, south and the Randstad (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 11: Central bank exposure and commercial banks' lending portfolio

	(1) Total lending	(2) Liquid lending	(3) Illiquid lending	(4) Repo lending	(5) Liquid assets
CB exposure * Post	-0.403**	-0.129	-0.222**	-0.052	-0.015
	(0.163)	(0.119)	(0.090)	(0.069)	(0.020)
# branches * Post	-0.003	-0.009	0.011	-0.004	0.002
	(0.017)	(0.006)	(0.013)	(0.004)	(0.004)
Log(population), yearly	3.964***	0.352	3.605***	0.006	-0.187
	(0.976)	(0.337)	(0.674)	(0.276)	(0.138)
Pre-shock mean	0.683	0.097	0.474	0.112	0.030
Post-shock mean	1.130	0.221	0.771	0.139	0.077
Observations	523	523	523	523	523
R-squared	0.682	0.631	0.745	0.679	0.634
Banks	70	70	70	70	70
Period			1910-1918		
Bank FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Region-Year FE	Y	Y	Y	Y	Y

Note: This table presents annual bank balance sheet items for 1910-1918. Post is a dummy that equals 1 if the year is 1914 or later. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All dependent variables are weighted by their 1913 (pre-war) total assets. Total lending the sum of all credit lines, repo lending, short term collateralized loans (bills of exchange) and long-term loans. Liquid lending consists of bills of exchange, while illiquid lending contains credit lines and long-term loans. Liquid assets include cash and cash equivalents and Dutch government bonds. All columns control for the log of the yearly population of the bank's pre-war office locations and for the size of its branch network interacted with Post. All columns include bank, year and region x year fixed effects, where the regions are north, middle, south and the Randstad (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table 12: Banks' balance sheets and central bank exposure: Heterogeneity by bank size

	(1) Total assets	(2) Deposits	(3) Equity	(4) Total lending	(5) Illiquid lending	(6) Repo lending
CB exposure * Post	-0.649**	-0.505**	-0.130*	-0.515**	-0.302**	-0.075
	(0.277)	(0.204)	(0.068)	(0.215)	(0.129)	(0.087)
Large bank * Post	-0.179	0.047	-0.035	0.093	-0.114	0.025
	(0.344)	(0.229)	(0.065)	(0.245)	(0.114)	(0.083)
CB exposure * Large bank * Post	0.477	0.108	0.027	0.139	0.192	0.026
	(0.411)	(0.245)	(0.095)	(0.292)	(0.161)	(0.092)
# branches * Post	0.005	-0.009	0.006	-0.015	0.010	-0.007
	(0.035)	(0.014)	(0.007)	(0.019)	(0.014)	(0.004)
Log(population), yearly	6.566***	7.235***	0.237	3.934***	3.593***	-0.003
	(1.601)	(1.343)	(0.290)	(0.964)	(0.670)	(0.279)
Observations	517	518	505	518	518	518
R-squared	0.666	0.788	0.634	0.673	0.734	0.680
Banks	68	68	67	68	68	68
Period				1910-1918		
Bank FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Region-Year FE	Y	Y	Y	Y	Y	Y

Note: This table presents annual bank balance sheet items for 1910-1918. Six banks in the balance sheet sample do not report deposits in more than one year and therefore drop out of the deposits column. Post is a dummy that equals 1 if the year is 1914 or later. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. Large bank is a dummy that equals 1 for banks above the median 1913 size. All dependent variables are weighted by their 1913 (pre-war) total assets. Total Assets is thus the total assets of a bank i in year t over total assets of 1913. Total deposits is the total deposits and current account outstanding balances, Equity is the sum of capital and retained earnings. Total lending the sum of all credit lines, repo lending, short term collateralized loans (bills of exchange) and long-term loans. Illiquid lending contains credit lines and long-term loans. All columns control for the log of the yearly population of the bank's pre-war office locations and for the size of its branch network interacted with Post. All columns include bank, year and region x year fixed effects, where the regions are north, middle, south and the Randstad (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

A Historical Background

A.1 De Nederlandsche Bank

DNB was established in 1814 via royal decree to "lift up commerce from its deplorable state [...] in order to be able to nourish a justified hope for the return of that prosperity which our fathers have enjoyed" (Kymmell, 1992, p. 61). That is, DNB's primary function was to boost trade by extending credit via discounting and collateralized lending. Contrary to most central banks at the time, DNB was not set up as a 'bank for the state': its charter did not allow for extending unsecured credit, not even to the government (Vanthoor, 2005). The charter lasted 25 years and was renewed several times until DNBs nationalization in 1948.

The expansion of DNB's branch network after 1864 boosted commercial credit provision across the country (Vanthoor, 2005). Financial institutions accepted to the discount window could expand their business and loosen their credit conditions. In most regional offices, cashiers, bankers and brokers would discount paper on behalf of firms and individuals, and Van der Werf (1988) even claims that the regional banking sector could not survive without DNB's discounting facilities. DNB was thus very much a complement rather than a substitute to the commercial banks. According to Kymmell (1992), DNB became the bankers' bank in the provinces after the opening of the branches. Indeed, DNB presidents consistently opposed attracting retail deposits from the banking sector. President Willem Mees (1863–1885) argued that deposit-taking was solely the business of banks and cashiers, and his successor Nicholaas Pierson (1885–1891) shared the view that it should never be the Bank's aim to take away deposits from other banks (Vanthoor, 2005).

Moreover, since its establishment DNB acted as the general lender to the financial system when it faced liquidity shortages (Vanthoor, 2005). It is not straightforward to pinpoint an exact moment when the Bank could be considered a lender of last resort, as it never had to perform the function widely before 1914. The secretary (and soon to be president) of DNB in 1863, Mees, mentions the role the bank played to prevent liquidity crises in the previous decade (Wijtvliet, 1993). Jonker (1996) states that DNB becomes more of lender of last resort when commercial banks started to emerge in the 1860s. Capie et al. (1994) argue that DNB adopted the lender of last resort role in the 1870s, while Mooij and Prast (2003) mention that while the Bank Act of 1903 did not explicitly mention it, DNB was de facto the lender of last resort. According to Vanthoor (2005), DNB had definitely lost its role as lender to trade and industry in 1914, laying the foundation for it's dual function as the national bankers' bank and lender of last resort.

DNB's retail deposit facilities also evolved over this period. In 1838, the renewal of its charter allowed retail depositing in a current account, but without the option to overdraft. From 1889, current account holders at DNB's head office (*Hoofdbank*) could overdraft their accounts, but only with appropriate collateral. In the branches, depositing in current account was also allowed from 1889 onward.

One of the reasons why DNB allowed for non-banks to deposit their money at its branches is because it wanted to modernize the payment system. Settling payments in the Netherlands was until 1900 still mostly cash. Making transfers from one place to another was expensive and slow: cheques were barely used because of stamp duty when sending it by post, and the large cities used different cashier's quittances as a means of payment (Vanthoor, 2005). DNB tried to make the payment system more efficient by allowing non-banks to open current accounts at its branches since 1889. The Bank envisioned that if enough residents would open an account at one of the DNB offices, a national payment network would make transfers significantly more efficient by simply reassigning balances on their branch books. However, multiple attempts to set up such a giro system failed due to a lack of interest and objection from the Amsterdam cashiers, where the payment system was concentrated thus far (Van Der Werf, 1988). The giro system was only rolled out in 1917.

A.2 Central bank retail deposits in other countries

Historically, the Netherlands was not the only country where the central bank accepted retail deposits. In 1876, the Reichsbank allowed all classes of the population "who enjoy the necessary confidence" to open accounts, though it required a minimum balance of 1,000 Marks (National Monetary Commission, 1910).[46] DNB, on the other hand, accepted any amount a current account holder wanted to deposit. At the Bank of England, any person that was "respectably introduced" could deposit moneys since 1695 (Antipa and Rieder, 2026). These retail deposits were widely used: retail deposits were on average more than a quarter of the Banking Department's total size. Similarly, Banco d'España even dominated the deposit-taking business in Spain around 1900 (**?**).

Contrary to DNB, the Bank of England and Banco d'España seem to have actively competed for deposits with the commercial banking sector. The incentives of the central bank to try and attract deposits are then arguably different, especially during crisis times. DNB presidents consistently maintained the view that deposit-taking was the business of commercial banks, not the central bank (Vanthoor, 2005).

[46] Comparable to *f*. 600 in 1914, which is around 9000 EUR today.

A.3 Commercial banks in the Netherlands

Commercial banks' deposits started increasing from 1890s onwards, when the banks actively tried to attract deposits by advertising and offering interest. Since 1875, commercial banks gradually adopted current accounts (credit lines) as one of the main forms of providing credit (Van Der Werf, 1988). Clients could also deposit money in the current accounts, and they functioned in a similar way as demandable deposits (and offered a comparable rate). This paper includes the liability side current account balances at commercial banks as a part of total bank deposits.

Despite this growth, the Netherlands had a low share of deposits as a percentage of the money supply M1 compared to other European countries: 34%, compared to 66% in Belgium, and 85% in the UK (Van Riel, 2021; Vanthoor, 2005). A reason for the relatively low level of deposits is the large money market existing at the time. Anyone could, via a bank or broker, invest in *prolongaties*, 1 month loans collateralized with any type of security listed on the Amsterdam Stock Exchange, with a haircut of minimum 10%. These repurchase agreements (repo)-like instruments were preferred to bank deposits since they provided a higher return than a 1 month bank deposit, while being considered as liquid and could be indefinitely rolled over (Van Ketel and Verbiest, 1923). The repo market amounted to $f.$ 400 million, or 15% of 1914 GDP (Van Riel, 2021).

The closure of the stock market in 1914-1915 led to a complete freeze of the repo market (see the next section), while commercial banks simultaneously received an inflow of deposits after the short-lived bank run subsided. Whether the increasing importance of deposits for banks' funding is caused by the repo market breakdown of 1914 cannot be causally identified. Few losses materialized after the stock market was reopened in February 1915, due to the increasing returns on mainly US stocks (Jongman, 1960). The repo market became more volatile but remained widely used until after World War II.

A.4 War measures by the government and DNB

After the outbreak of WWI on 28th of July, the Dutch government implemented several emergency measures: closing the stock exchange, initiating general mobilization and, most importantly, suspending the Gold Standard. On 3 August, the official metal-cover ratio was by law allowed to deviated from 40 to 20 percent, thus officially suspending gold convertibility.[47] The demand for banknotes and especially silver by the public was unprecedented (gold exports were prohibited) and could not be met with the available minted silver coins. The government therefore authorized the issuance of up to $f.$ 25

[47]The metal-cover ratio (*metaaldekking*) was by law set to 40%, meaning DNB had to hold 40% of its demandable debt (banknotes in circulation and current accounts) in gold.

million in "zilverbons" (silver certificates) to counteract silver outflows. These certificates were a temporary means of payment that could be exchanged to silver in the future (Vanthoor, 2005).[48] By October 1914, the metal-cover ratio had fallen to its lowest point of 32.59 percent before stabilizing.

DNB set up support syndicate with a large group of commercial banks and bankers' houses to provide liquidity to those affected by the closure of the Stock Exchange. Many brokers, dealers and banks had substantial volume of repo contracts (*prolongaties*) outstanding before the war: around $f.$ 400 million, which was twice the size of term deposits by banks (Van Riel, 2021). The entire repo market froze once the Stock Exchange closed, leaving many investors short of liquidity. The support syndicate was initially $f.$ 25 million with the idea that the Stock Exchange would reopen the day after, but was quickly increased to $f.$ 200 million when the reopening was postponed.[49] DNB was responsible for the first $f.$ 40 million of the syndicate, and above that commercial banks would guarantee 20% of the additional loans required by the market, up to the maximum of their voluntary commitment.

[48]The zilverbons had to be withdrawn as soon as DNB had enough silver coins available to the government. The certificates were given legal tender and were very popular: the maximum amount to be issued expanded to $f.$ 40 million in 1916 and $f.$ 100 million in 1918 (Vanthoor, 2005).

[49]Eventually, the stock market reopened on February 9, 1915.

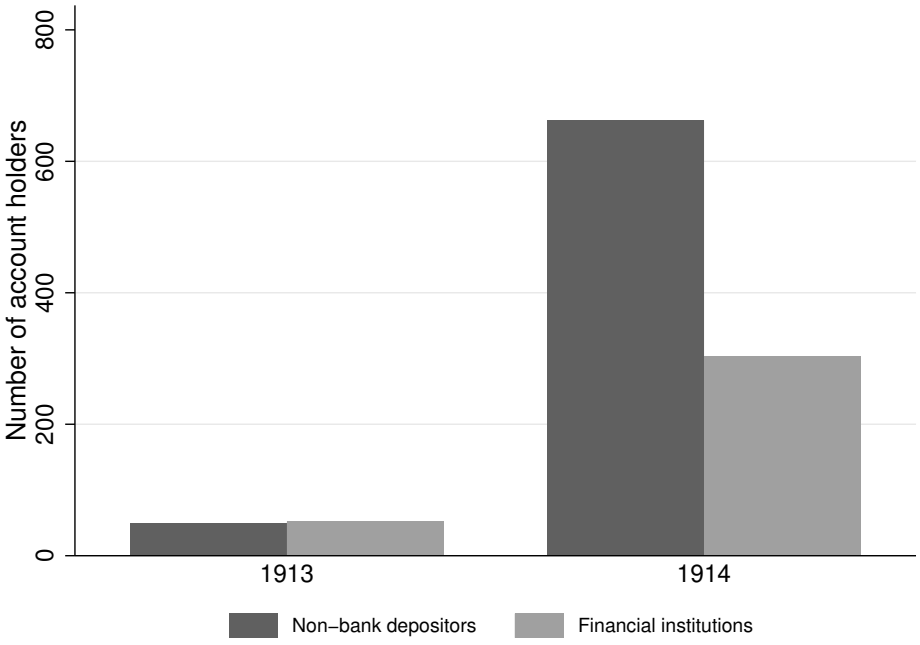
B Figures

Figure B.1: DNB Branch Network in 1914

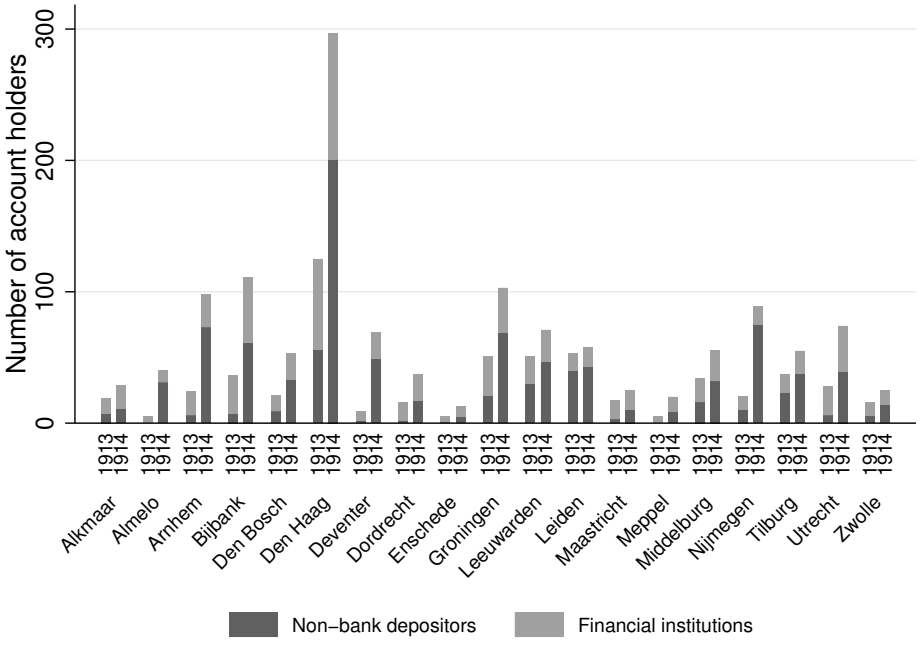


Note: This map shows all the DNB branches in 1914 (circled in blue), and their corresponding areas (bordered in red). Source: NA 2.25.68, 16158: "Vaststelling en wijziging van de ressorten van Bijbank en agentschappen, de grenzen daarvan en de daarbinnen vallende correspondentschappen 1890 - 1981." Red drawn borders are author's own adjustments to 1914 situation.

Figure B.2: DNB current account holders by branch



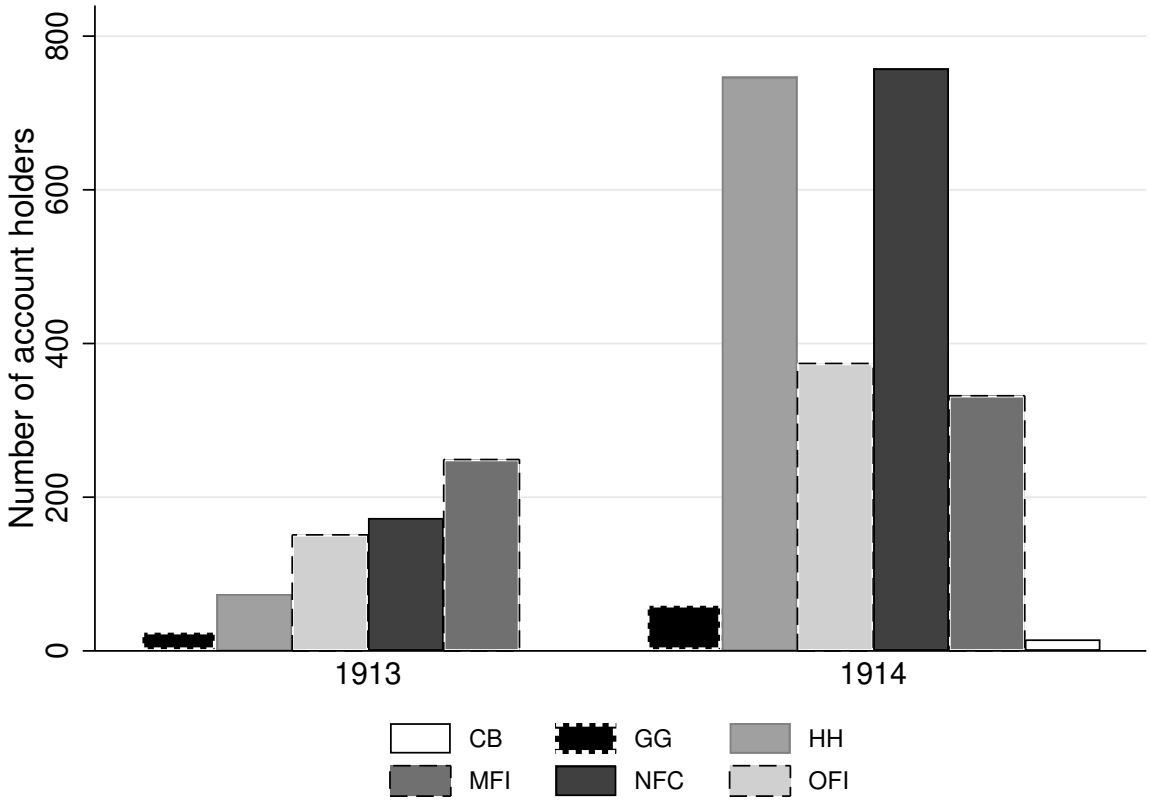
(a) Amsterdam Head Office



(b) DNB branches

Note: Figure (a) shows the count and composition of the current account holders of DNB at the head office in Amsterdam in 1913 and 1914. Figure (b) shows the count and composition of the current account holders at all DNB branches in 1913 and 1914. Names are digitized from DNB's yearly list of account holders and manually matched to the Financial Address 1914 yearbook to identify financial institutions. DNB Namelist of current account holders, NA 2.25.68 - 16431; Financieel Adresboek 1914.

Figure B.3: DNB current account holders by sector



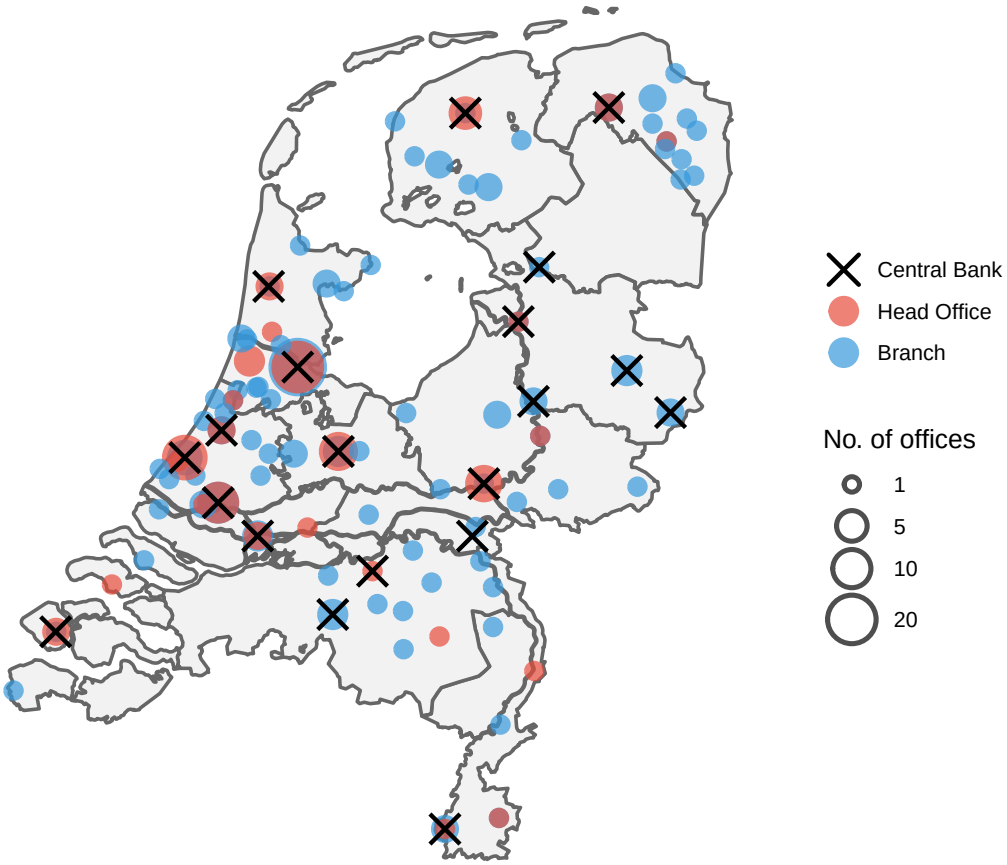
Note: this figure shows the count and composition of the current account holders at all DNB branches in 1913 and 1914 into sectors. CB covers all central bank (sub)branches and correspondent offices, MFI include all banks that accepted deposits, OFI include brokers, insurance companies, real estate investment companies, and other institutions that appear in the Financial Addressbook of 1914. GG is national and local governments, NFC are all firms, and HH are all individuals.. Names are digitized from DNB's yearly list of account holders and manually matched to the Financial Address 1914 yearbook to identify financial institutions. DNB Namelist of current account holders, NA 2.25.68 - 16431; Financieel Adresboek 1914.

Figure B.4: Excerpt from bank advertisements in Algemeen Handelsblad



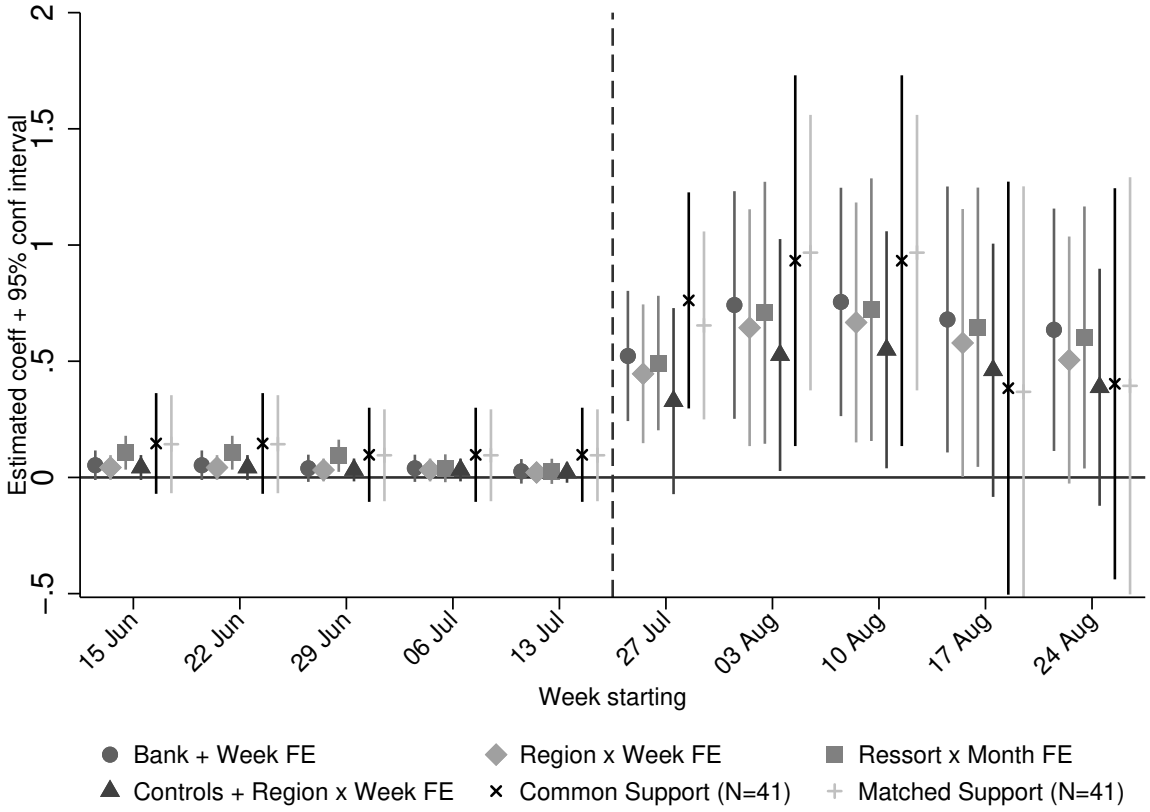
Note: this is a screenshot from the Dutch national newspaper Algemeen Handelsblad from September 1, 1914. The newspaper can be accessed via Delpher (https://www.delpher.nl/nl/kranten) using the following search terms: "Twentsche" AND "deposito".

Figure B.5: Geographical variation of bank branches, banks in balance sheet sample



Note: This map shows the location of all commercial bank and central bank branches. The black crosses indicate the 20 central bank branches, the red dots are the locations of head offices of the commercial banks, and the blue dots are the location of commercial bank branches. Note that one dot can contain more than one bank located in the respective city. Includes all banks of the balance sheet sample. Source: Financieel Adresboek 1914.

Figure B.6: Robustness in parallel trends: weekly coefficients are not significantly different before the shock

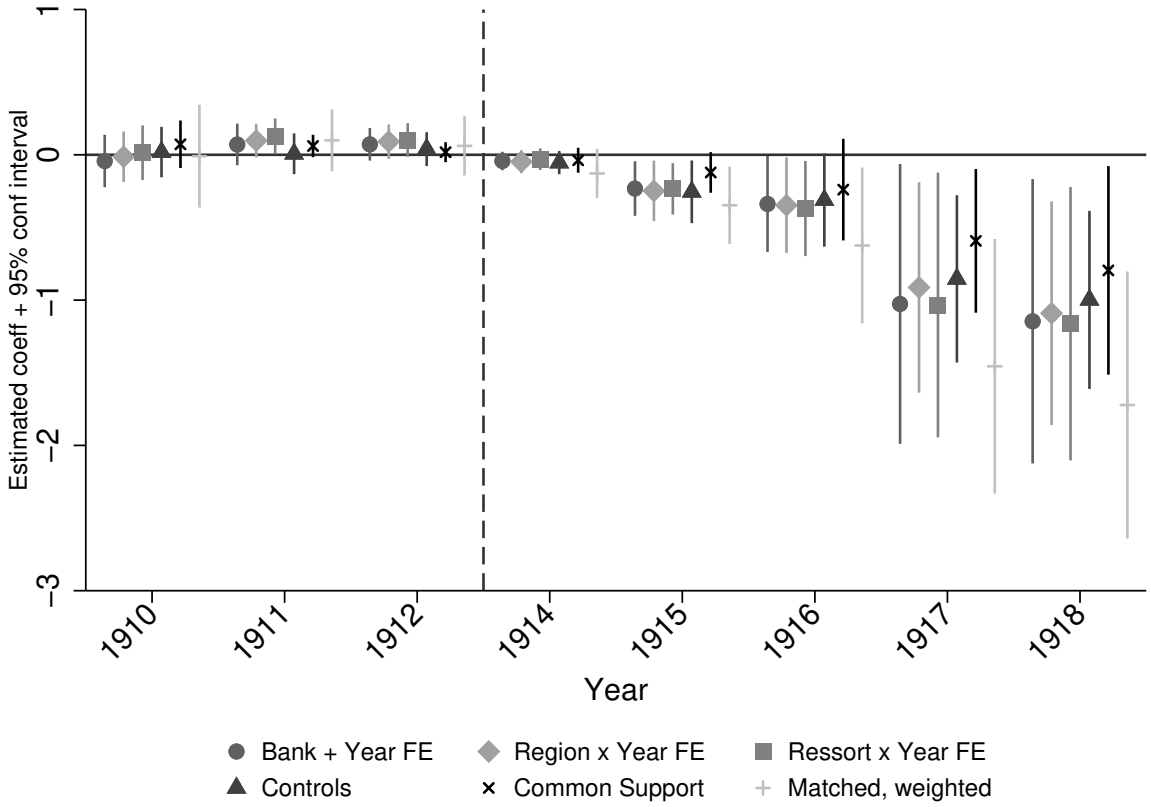


Note: This figure shows β coefficients of the following specification:

$$Y_{i,t} = \sum_{t=1914w25}^{t=1914w35} \beta_t \; CB \; Exposure_i \times D_{week=t} + \gamma Branches_{i,1913} \times Post_t + \Gamma' X_{i,1909} \times Post_t + \kappa_i + \eta_t + \epsilon_{i,t} \tag{2}$$

Where $Y_{i,t}$ is the short-term deposit rate of bank i in week t, $CB\ Exposure_i$ is the fraction of bank's i branches that are located in a city that has a central bank office, $D_{week=t}$ is a dummy equal to 1 for week t, where week 30 (the last week before the shock takes place) is left out to avoid the dummy variable trap. The period covered is from June – August 1914. Bank and week fixed effects are included in all specifications, as well as the control variable Branches_i, which is the number of offices bank i operated before the shock. Both CB exposure_i and Branches_i are time-invariant and are therefore absorbed by the bank fixed effects α_i; only their interactions with Post_t are identified. Standard errors are clustered on the bank level. The other specifications include region × week fixed effects (diamond), central bank ressort × month fixed effects (square), or the region × week fixed effects plus additional control variables population, number of competing banks and unemployment rate (triangle). Two other specifications have a reduced sample of 41 banks: the common support (cross), comparing only banks that are similar in terms of city size, and matched support (plus), comparing treated banks only to the nearest 10 banks in terms of city size. The short-term commercial bank deposit rate is the simple average of the shortest rate offered by each commercial bank in the sample (only if a banks has demandable, daily or weekly). The vertical line reflects the outbreak of the war (28th of July 1914). Source: Delpher Digital Newspaper Archive, author's own computations.

Figure B.7: Robustness in parallel trends: yearly coefficient for banks' deposits

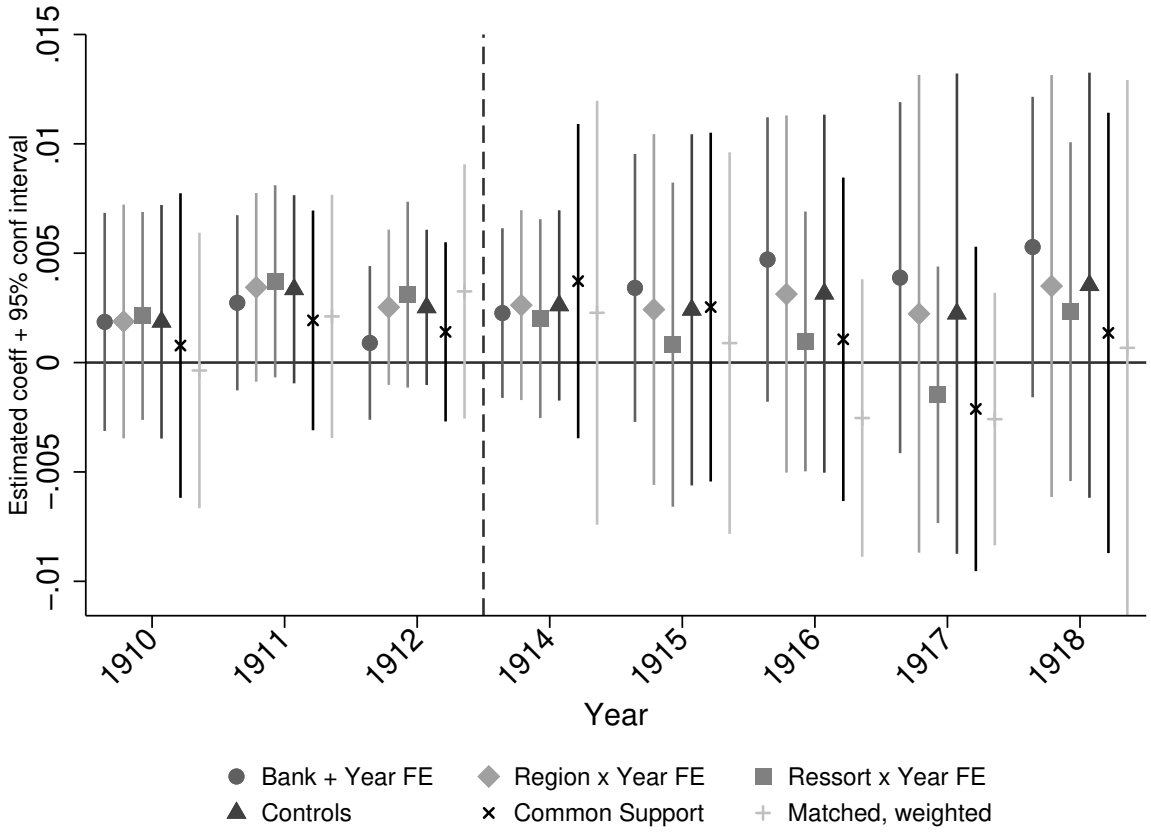


Note: This figure shows β coefficients of the following specification:

$$Y_{i,t} = \sum_{t=1910}^{t=1918} \beta_t \; CB \; Exposure_i \times D_{year=t} + \gamma Branches_{i,1913} \times Post_t + \Gamma_1' X_{i,1909} \times Post_t + \kappa_i + \eta_t + \epsilon_{i,t} \tag{4}$$

Where $Y_{i,t}$ is deposits of bank i in year t weighted by pre-war total assets (1913), $CB \; Exposure_i$ is the fraction of bank's i branches that are located in a city that has a central bank office, $D_{year=t}$ is a dummy equal to 1 for year t, where 1913 (the last year before the shock takes place) is left out to avoid the dummy variable trap. The period covered is 1910-1918. Bank and year fixed effects are included in all specifications, as well as the control variable Branches$_i$, which is the number of offices bank i operated before the shock. Both CB exposure$_i$ and Branches$_i$ are time-invariant and are therefore absorbed by the bank fixed effects α_i; only their interactions with Post$_t$ are identified. Standard errors are clustered on the bank level. The other specifications include region × year fixed effects (diamond), central bank ressort × year fixed effects (square), plus additional control variable yearly population served of bank i. Two other specifications have a reduced sample of 51 banks: the common support (cross), comparing only banks that are similar in terms of city size, and matched support (plus), comparing treated banks only to the nearest 10 banks in terms of city size. The vertical dashed line reflects the year of the shock, 1914.

Figure B.8: Stable yearly coefficients for banks' net interest margins



Note: This figure shows β coefficients of the following specification:

$$Y_{i,t} = \sum_{t=1910}^{t=1918} \beta_t \; CB \; Exposure_i \times D_{year=t} + \gamma Branches_{i,1913} \times Post_t + \Gamma_1' X_{i,1909} \times Post_t + \kappa_i + \eta_t + \epsilon_{i,t} \tag{6}$$

Where $Y_{i,t}$ is the net interest margin of bank i in year t weighted by current total assets, $CB\ Exposure_i$ is the fraction of bank's i branches that are located in a city that has a central bank office, $D_{year=t}$ is a dummy equal to 1 for year t, where 1913 (the last year before the shock takes place) is left out to avoid the dummy variable trap. The period covered is 1910-1918. Bank and year fixed effects are included in all specifications, as well as the control variable Branches_i, which is the number of offices bank i operated before the shock. Both CB exposure_i and Branches_i are time-invariant and are therefore absorbed by the bank fixed effects α_i; only their interactions with Post_t are identified. Standard errors are clustered on the bank level. The other specifications include region $\times$ year fixed effects (diamond), central bank ressort $\times$ year fixed effects (square), plus additional control variable yearly population served of bank i. Two other specifications have a reduced sample of 51 banks: the common support (cross), comparing only banks that are similar in terms of city size, and matched support (plus), comparing treated banks only to the nearest 10 banks in terms of city size. The vertical dashed line reflects the year of the shock, 1914.

C Tables

Table C.1: Short-term deposit rates, inverted term structure and central bank exposure: longer windows

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Demandable deposit rate				Inverted term structure			
Post	0.440***				0.067			
	(0.094)				(0.067)			
CB exposure * Post	-0.127	0.202	0.206	0.199	0.055	0.132*	0.157**	0.108
	(0.114)	(0.129)	(0.139)	(0.154)	(0.085)	(0.068)	(0.078)	(0.071)
Observations	7,414	3,702	3,702	3,702	7,444	3,731	3,731	3,731
R-squared	0.634	0.687	0.707	0.733	0.412	0.558	0.570	0.593
Within R-sq	0.058	0.009	0.010	0.008	0.034	0.014	0.019	0.008
Banks	76	76	76	76	77	77	77	77
Period	1914-1915		1914		1914-1915		1914	
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	N	N	N	Y	N	N	N
Week FE	N	Y	Y	Y	N	Y	Y	Y
Region-Month FE	N	N	Y	N	N	N	Y	N
Ressort-Month FE	N	N	N	Y	N	N	N	Y

Note: This table repeats the estimates of Tables 3 and 6 over longer windows. In columns (1)-(4) the dependent variable is the shortest deposit rate available for each bank (max weekly rate); in columns (5)-(8) it is the inverted term structure, a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable rate is higher than weekly rate and/or weekly rate is higher than monthly, or demandable rate is higher than monthly rate), and 0 otherwise; banks that quote only one of these tenors are coded 0. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. Columns (1) and (5) cover 1914-1915 and include year fixed effects. Columns (2)-(4) and (6)-(8) include all 1914 observations and week fixed effects. Columns (3) and (7) also include region x month fixed effects, where the regions are north, middle, south and 'Randstad' (North- and South-Holland). Columns (4) and (8) include central bank office region (see Figure B.1) x month fixed effects. All columns include bank fixed effects. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.2: Central bank exposure and World War I exposure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Short-term rate				Inverted term structure		
CB exposure * Post	0.212	0.444**	0.388*	0.424**	0.150**	0.187***	0.203***	0.170**
	(0.140)	(0.208)	(0.199)	(0.211)	(0.075)	(0.068)	(0.072)	(0.069)
War exposed * Post	0.075	-0.114	0.166	0.064	-0.087	-0.037	0.020	0.023
	(0.129)	(0.196)	(0.159)	(0.171)	(0.060)	(0.086)	(0.101)	(0.123)
Observations	3,702	640	640	640	3,731	643	643	643
R-squared	0.707	0.611	0.637	0.669	0.572	0.604	0.612	0.625
Within R-sq	0.011	0.027	0.021	0.022	0.024	0.022	0.023	0.015
Banks	76	76	76	76	77	77	77	77
Period	1914		Jul-Aug		1914		Jul-Aug	
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y	Y	Y
Region-Month FE	Y	N	Y	N	Y	N	Y	N
Ressort-Month FE	N	N	N	Y	N	N	N	Y

Note: This table presents estimates for demandable deposit rates (1)-(4), and inverted term structure of deposit pricing by commercial banks (5)-(8). Demandable rate includes the shortest deposit rate available for each bank (max weekly rate). Inverted term structure is a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable rate higher than the weekly or monthly rate, or weekly rate higher than the monthly rate), and 0 otherwise. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. War exposure is a dummy that equals 1 if a bank has at least one branch in a province that borders one of the countries at war, and zero otherwise. All columns include week and bank fixed effects. Columns (1) and (5) include all 1914 observations, Columns (2)-(4) and (6)-(8) include only observations in the month before (July) and the month after (August) the shock. Columns (1), (3), (5), and (7) also include region-time fixed effects, where the regions are north, middle, south and 'Randstad' (North- and South-Holland). Columns (4) and (8) include central bank office region x month fixed effects. (central bank office region (see Figure A.1) x month). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.3: Central bank exposure and regional controls, inverted deposit pricing

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Inverted term structure of deposit pricing									
CB exposure * Post	0.157**	0.170**	0.141	0.140	0.154	0.201***	0.261***	0.148	0.146	0.156
	(0.078)	(0.081)	(0.095)	(0.095)	(0.095)	(0.071)	(0.079)	(0.103)	(0.105)	(0.103)
# branches * Post		0.004	-0.000	-0.000	-0.007		0.021*	0.005	0.005	-0.006
		(0.007)	(0.009)	(0.009)	(0.012)		(0.012)	(0.013)	(0.014)	(0.016)
Log(population) * Post			0.018	0.019	0.064			0.064*	0.065*	0.135**
			(0.033)	(0.033)	(0.055)			(0.033)	(0.033)	(0.062)
Unemployment rate * Post				0.002	-0.014				0.012	-0.011
				(0.022)	(0.017)				(0.023)	(0.018)
# Banks * Post					-0.004					-0.006
					(0.003)					(0.003)
Observations	3,731	3,731	3,731	3,731	3,731	643	643	643	643	643
R-squared	0.570	0.570	0.571	0.571	0.574	0.612	0.618	0.624	0.625	0.629
Within R-sq	0.019	0.020	0.021	0.021	0.028	0.022	0.038	0.053	0.055	0.066
Banks	77	77	77	77	77	77	77	77	77	77
Period			1914					Jul-Aug		
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Region-Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: This table presents estimates for inverted term structure of deposit rates by commercial banks including regional controls. Inverted term structure is a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable rate is higher than weekly rate and/or weekly rate is higher than monthly, or demandable rate is higher than monthly rate), and 0 otherwise. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. Columns (1)-(5) cover 1914, while columns (6)-(10) include only observations in the month before (July) and the month after (August) the shock. # branches is the number of offices a bank has pre-shock. This excludes branches outside of the Netherlands (3 banks have an office in London or Batavia). Log(population) is the natural logarithm of the total population that a bank serves, i.e. the sum of all city inhabitants of every branch location. Unemployment rate is the population-weighted average percentage unemployed across a banks' branches. # Banks is the average number of banks (population-weighted) in its branch locations. All specifications include bank, week and region-month fixed effects, where the regions are north, middle, south and 'Randstad' (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.4: Central bank exposure and exposure to Amsterdam

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Demandable deposit rate					Inverted term structure				
		All		No AMS	All	No AMS		All		No AMS	All	No AMS
CB exposure	0.204				0.126		-0.009				-0.006	
	(0.160)				(0.193)		(0.046)				(0.046)	
Amsterdam	-0.334*				-0.295*		0.032				-0.085	
	(0.172)				(0.167)		(0.115)				(0.092)	
CB exposure * Post	0.185	0.360***	0.378***		0.405**	0.394*	0.130*	0.184**	0.164*		0.235***	0.223**
	(0.152)	(0.135)	(0.134)		(0.174)	(0.201)	(0.072)	(0.085)	(0.094)		(0.077)	(0.085)
Amsterdam * Post		0.772						1.019				
		(1.373)						(1.118)				
CB exposure * Amsterdam * Post		-0.778						-1.055				
		(1.428)						(1.141)				
# branches * Post	0.033*	0.048***	0.043***		0.044**	0.043*	0.004	0.003	-0.001		0.021*	0.002
	(0.017)	(0.014)	(0.015)		(0.018)	(0.023)	(0.007)	(0.007)	(0.006)		(0.012)	(0.008)
Observations	3,702	3,702	2,662		640	460	3,731	3,731	2,691		643	463
R-squared	0.272	0.718	0.756		0.316	0.659	0.067	0.572	0.594		0.127	0.641
Within R-sq	0.060	0.046	0.041		0.082	0.023	0.011	0.024	0.030		0.046	0.036
Banks	76	76	56		76	56	77	77	57		77	57
Period		1914			Jul-Aug			1914			Jul-Aug	
Bank FE	N	Y	Y		N	Y	N	Y	Y		N	Y
Week FE	Y	Y	Y		Y	Y	Y	Y	Y		Y	Y
Region-Month FE	Y	Y	Y		Y	Y	Y	Y	Y		Y	Y

Note: This table presents estimates for demandable deposit rates (1)-(5) and inverted deposit pricing by commercial banks (6)-(10). Columns (1)-(3) and (6)-(8) cover 1914; columns (4), (5), (9) and (10) include only the month before (July) and the month after (August) the shock. Columns (3), (5), (8) and (10) exclude any bank that has exposure to Amsterdam, i.e. where at least one branch is located in the city. Demandable rate includes the shortest deposit rate available for each bank (max weekly rate). Inverted term structure is a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable higher than weekly and weekly higher than monthly), and 0 otherwise. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous fraction of branch overlap between a commercial bank and the central bank. Amsterdam exposure is the share of commercial bank branches that is located in Amsterdam. All columns include week and region-month (north, south, middle and 'Randstad' (North- and South-Holland)) fixed effects. All specifications also control for the size of a bank's branch network interacted with Post, so that the estimated effect of central bank exposure is not picking up the difference between a one-branch and a ten-branch bank. The number of branches is time-invariant and is absorbed by the bank fixed effects wherever those are included. The levels of CB exposure and Amsterdam exposure are time-invariant and are therefore only identified in the columns without bank fixed effects. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.5: Central bank exposure and exposure to the largest cities

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Demandable deposit rate					Inverted term structure				
	All		Without B4	All	Without B4	All		Without B4	All	Without B4
CB exposure	0.166			0.050		-0.066			-0.078	
	(0.181)			(0.209)		(0.054)			(0.063)	
Large city	-0.082			0.018		0.136*			0.114	
	(0.194)			(0.204)		(0.080)			(0.099)	
CB exposure * Post	0.180	0.417***	0.383**	0.393**	0.220	0.122*	0.220**	0.210*	0.223***	0.116
	(0.155)	(0.135)	(0.151)	(0.178)	(0.156)	(0.072)	(0.098)	(0.121)	(0.077)	(0.096)
Large city * Post		-0.195					0.642			
		(0.676)					(0.521)			
CB exposure * Large city * Post		0.079					-0.677			
		(0.678)					(0.519)			
# branches * Post	0.033*	0.048***	0.040**	0.043**	0.040*	0.003	0.002	0.000	0.020	0.000
	(0.017)	(0.014)	(0.016)	(0.018)	(0.023)	(0.007)	(0.008)	(0.007)	(0.012)	(0.004)
Observations	3,702	3,702	1,612	640	279	3,731	3,731	1,641	643	282
R-squared	0.252	0.718	0.768	0.305	0.719	0.080	0.574	0.637	0.130	0.705
Within R-sq	0.036	0.047	0.047	0.066	0.022	0.025	0.029	0.071	0.048	0.023
Banks	76	76	35	76	35	77	77	36	77	36
Period		1914			Jul-Aug		1914			Jul-Aug
Bank FE	N	Y	Y	N	Y	N	Y	Y	N	Y
Week FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Region-Month FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

Note: This table presents estimates for demandable deposit rates (1)-(5) and inverted deposit pricing by commercial banks (6)-(10), with and without exposure to the largest cities. Columns (1)-(3) and (6)-(8) cover 1914; columns (4), (5), (9) and (10) include only the month before (July) and the month after (August) the shock. Columns (3), (5), (8) and (10) exclude any bank that has exposure to the largest four cities (B4), i.e. any branch located in Amsterdam, Rotterdam, The Hague or Utrecht (which all have above 100k inhabitants). Demandable rate includes the shortest deposit rate available for each bank (max weekly rate). Inverted term structure is a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable higher than weekly and weekly higher than monthly), and 0 otherwise. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous fraction of branch overlap between a commercial bank and the central bank. Large city is the share of branches that is located in one of the largest cities. All columns include week and region-month (north, south, middle and 'Randstad' (North- and South-Holland)) fixed effects. All specifications also control for the size of a bank's branch network interacted with Post, so that the estimated effect of central bank exposure is not picking up the difference between a one-branch and a ten-branch bank. The number of branches is time-invariant and is absorbed by the bank fixed effects wherever those are included. The levels of CB exposure and large-city exposure are time-invariant and are therefore only identified in the columns without bank fixed effects. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.6: Central bank exposure and bank branches

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Demandable deposit rate				Inverted term structure			
	All		Unit	Multi-branch	All		Unit	Multi-branch
CB exposure	0.127				0.007			
	(0.186)				(0.039)			
# branches	0.050***				0.016*			
	(0.017)				(0.009)			
CB exposure * Post	0.283	0.551**	0.701***	0.289	0.179**	0.121	0.226**	0.236**
	(0.182)	(0.210)	(0.143)	(0.346)	(0.068)	(0.102)	(0.097)	(0.112)
# branches * Post		0.060		0.078***		-0.011		0.033*
		(0.043)		(0.025)		(0.014)		(0.016)
CB exposure * # branches * Post		-0.004				0.044		
		(0.051)				(0.028)		
Observations	640	640	351	289	643	643	354	289
R-squared	0.299	0.646	0.637	0.658	0.116	0.621	0.595	0.671
Within R-sq	0.058	0.043	0.041	0.070	0.033	0.046	0.019	0.079
Banks	76	76	42	34	77	77	43	34
Period	Jul-Aug							
Bank FE	N	Y	Y	Y	N	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y	Y	Y
Region-Month FE	Y	Y	Y	Y	Y	Y	Y	Y

Note: This table presents estimates for demandable deposit rates (1)-(4) and inverted deposit pricing by commercial banks (5)-(8) over the period July-August 1914, for banks with single and multi-branch networks. Columns (3) and (7) include only unit banks, i.e. commercial banks with only one branch, while columns (4) and (8) include only multi-branch banks. Demandable rate includes the shortest deposit rate available for each bank (max weekly rate). Inverted term structure is a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable higher than weekly and weekly higher than monthly), and 0 otherwise. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous fraction of branch overlap between a commercial bank and the central bank. # branches is the number of branches that a commercial bank has pre-war. Branches outside of the Netherlands are excluded from the count (3 banks have an office in London or Batavia). All columns include week and region-month (north, south, middle and 'Randstad' (North- and South-Holland)) fixed effects. Columns (4) and (8) additionally control for the number of branches interacted with Post; columns (3) and (7) hold the number of branches fixed by construction, since every bank in them has exactly one branch. The levels of CB exposure and the number of branches are time-invariant and are therefore only identified in the columns without bank fixed effects. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.7: Binary central bank exposure and commercial bank deposit rates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Demandable rate				Inverted term structure		
CB exposure * Post	0.125	0.166	0.345**	0.242*	0.090	0.049	0.102	0.075
	(0.101)	(0.150)	(0.143)	(0.143)	(0.065)	(0.083)	(0.083)	(0.095)
# branches * Post			0.061***	0.052***			0.018	0.016
			(0.017)	(0.016)			(0.012)	(0.012)
# Banks * Post				0.008*				0.002
				(0.004)				(0.002)
Observations	3,702	640	640	640	3,731	643	643	643
R-squared	0.706	0.632	0.642	0.649	0.566	0.604	0.608	0.610
Within R-sq	0.006	0.006	0.033	0.051	0.011	0.002	0.013	0.018
Banks	76	76	76	76	77	77	77	77
Period	1914		Jul-Aug		1914		Jul-Aug	
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y	Y	Y
Region-Month FE	Y	Y	Y	Y	Y	Y	Y	Y

Note: This table presents estimates for demandable deposit rates (1)-(4) and inverted term structure of deposit pricing by commercial banks (5)-(8), over 1914 in columns (1) and (5) and over July-August 1914 in columns (2)-(4) and (6)-(8). Demandable rate includes the shortest deposit rate available for each bank (max weekly rate). Inverted term structure is a dummy that equals 1 if the remuneration is higher for shorter-maturity deposits (demandable higher than weekly and weekly higher than monthly), and 0 otherwise. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure here is a dummy that equals 1 if a bank is fully exposed to the central bank, i.e. all its branches are located in CB cities, and zero otherwise. # branches is the number of branches that a commercial bank has pre-war. Branches outside of the Netherlands are excluded from the count (3 banks have an office in London or Batavia). # Banks is the average number of banks (population-weighted) in its branch locations. All columns include week, bank and region-month (north, middle, south and 'Randstad' (North- and South-Holland)) fixed effects. Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.8: Central bank exposure and deposit pricing under region x week fixed effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Demandable rate				Inverted term structure			
CB exposure * Post	0.201	0.425***	0.336	0.471**	0.155*	0.168*	0.193**	0.201*
	(0.142)	(0.130)	(0.208)	(0.185)	(0.079)	(0.098)	(0.075)	(0.111)
# branches * Post		0.067***		0.049**		-0.005		-0.002
		(0.017)		(0.022)		(0.012)		(0.016)
Log(population) * Post		-0.091		0.050		0.058		0.125*
		(0.079)		(0.105)		(0.058)		(0.066)
Unemployment rate * Post		0.010		-0.087*		-0.018		-0.021
		(0.030)		(0.048)		(0.017)		(0.019)
# Banks * Post		0.003		-0.002		-0.004		-0.006*
		(0.005)		(0.007)		(0.003)		(0.004)
Observations	3,702	3,702	640	640	3,731	3,731	643	643
R-squared	0.710	0.722	0.646	0.661	0.573	0.577	0.622	0.636
Within R-sq	0.009	0.050	0.014	0.055	0.018	0.028	0.020	0.059
Banks	76	76	76	76	77	77	77	77
Period	1914		Jul-Aug		1914		Jul-Aug	
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y
Week FE	Y	Y	Y	Y	Y	Y	Y	Y
Region-Week FE	Y	Y	Y	Y	Y	Y	Y	Y

Note: This table repeats the baseline specifications with region x week instead of region x month fixed effects. The dependent variable is the shortest deposit rate available for each bank (max weekly rate) in columns (1)-(4) and the inverted term structure dummy in columns (5)-(8). Columns (1), (2), (5) and (6) cover 1914; columns (3), (4), (7) and (8) include only the month before (July) and the month after (August) the shock. Columns (2), (4), (6) and (8) add the number of branches, the log of the population served, the population-weighted unemployment rate and the population-weighted number of banks in a bank's branch locations, each interacted with Post. Post is a dummy that equals 1 if the period of observation is after 26th of July 1914. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. Region x week fixed effects are the finer counterpart of the region x month fixed effects used in the main tables, and match the granularity of the weekly observations; they nest the week fixed effects. The regions are north, middle, south and 'Randstad' (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.9: Central bank exposure, bank size and funding composition

	(1)	(2)	(3)	(4)	(5)	(6)
	Size	Share of total assets				
	Log(total assets)	Deposits	Secured funding	Unsecured funding	Equity	Other liabilities
CB exposure * Post	-0.228*	-0.094**	0.023	0.036	0.003	0.032
	(0.134)	(0.036)	(0.037)	(0.030)	(0.035)	(0.033)
# branches * Post	0.013	0.003	-0.005	-0.000	0.001	0.002
	(0.017)	(0.005)	(0.005)	(0.002)	(0.004)	(0.005)
Log(population), yearly	1.857***	0.650***	0.257	-0.445***	-0.304***	-0.158
	(0.700)	(0.205)	(0.194)	(0.129)	(0.089)	(0.133)
Pre-shock mean	15.207	0.309	0.134	0.224	0.284	0.049
Post-shock mean	15.601	0.404	0.076	0.184	0.255	0.081
Observations	517	517	517	517	517	517
R-squared	0.975	0.878	0.788	0.899	0.790	0.704
Banks	71	71	71	71	71	71
Period	1910-1918					
Bank FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Region-Year FE	Y	Y	Y	Y	Y	Y

Note: This table presents annual bank balance sheet items for 1910-1918. Six banks in the balance sheet sample do not report deposits in more than one year and are therefore excluded from all columns. Column (1) is the natural log of total assets. Columns (2)-(6) divide each liability item by the bank's total assets in the same year; Other liabilities is the residual, so the five shares sum to one and their coefficients sum to zero. All columns use the same sample of bank-years in which deposits, equity and total assets are observed. Post is a dummy that equals 1 if the year is 1914 or later. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. Total deposits is the total deposits and current account outstanding balances, Secured funding is all short-term collateralized borrowing (repo and bills of exchange), Unsecured funding is the sum of interbank borrowing, creditors and long-term borrowing, Equity is the sum of capital and retained earnings. All columns control for the log of the yearly population of the bank's pre-war office locations and for the size of its branch network interacted with Post. All columns include bank, year and region x year fixed effects, where the regions are north, middle, south and the Randstad (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. Wild cluster bootstrap p-values (Rademacher weights, 9,999 replications) for CB exposure x Post, by column: (1) 0.112; (2) 0.016; (3) 0.573; (4) 0.251; (5) 0.945; (6) 0.335. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.10: Central bank exposure and bank balance sheets: full sample and common support

	(1)	(2)	(3)	(4)	(5)	(6)
	Full sample			Common support		
	Total assets	Deposits	Total lending	Total assets	Deposits	Total lending
CB exposure * Post	-0.414*	-0.426***	-0.403**	-0.450*	-0.294***	-0.563***
	(0.224)	(0.147)	(0.163)	(0.224)	(0.104)	(0.130)
# branches * Post	0.014	-0.002	-0.003	0.015	0.003	-0.011
	(0.031)	(0.011)	(0.017)	(0.031)	(0.008)	(0.016)
Log(population), yearly	6.570***	7.261***	3.964***	-6.237*	-0.204	-2.541
	(1.650)	(1.354)	(0.976)	(3.477)	(1.601)	(2.478)
Pre-shock mean	0.943	0.306	0.683	0.944	0.275	0.711
Post-shock mean	1.617	0.681	1.130	1.600	0.560	1.134
Observations	522	523	523	364	364	364
R-squared	0.670	0.786	0.682	0.623	0.720	0.621
Banks	70	70	70	48	48	48
Period	1910-1918					
Bank FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Region-Year FE	Y	Y	Y	Y	Y	Y

Note: This table presents annual bank balance sheet items for 1910-1918. Six banks in the balance sheet sample do not report deposits in more than one year and therefore drop out of the deposits columns. Columns (1)-(3) use the full balance sheet sample; columns (4)-(6) include only banks in the common support of average market population, i.e. banks whose log average population across their branch locations lies within the range covered by both high- and low-exposed banks. All dependent variables are weighted by their 1913 (pre-war) total assets. Total lending is the sum of all credit lines, repo lending, short term collateralized loans (bills of exchange) and long-term loans. Post is a dummy that equals 1 if the year is 1914 or later. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All columns control for the log of the yearly population of the bank's pre-war office locations and for the size of its branch network interacted with Post. All columns include bank, year and region x year fixed effects, where the regions are north, middle, south and the Randstad (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.11: Central bank exposure and bank funding: adding control variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total assets				Deposits			
CB exposure * Post	-0.477**	-0.414*	-0.463*	-0.281	-0.476***	-0.426***	-0.463***	-0.299**
	(0.217)	(0.224)	(0.240)	(0.303)	(0.160)	(0.147)	(0.117)	(0.132)
# branches * Post		0.014	0.011	0.016		-0.002	-0.004	0.000
		(0.031)	(0.030)	(0.031)		(0.011)	(0.012)	(0.012)
Log(population), yearly		6.570***	6.614***	6.916***		7.261***	7.294***	7.583***
		(1.650)	(1.583)	(1.433)		(1.354)	(1.305)	(1.155)
Unemployment rate * Post			-0.021	-0.048			-0.016	-0.041
			(0.056)	(0.052)			(0.033)	(0.031)
# Banks * Post				-0.008				-0.007***
				(0.005)				(0.002)
Pre-shock mean	0.943	0.943	0.943	0.943	0.306	0.306	0.306	0.306
Observations	522	522	522	522	523	523	523	523
R-squared	0.631	0.670	0.671	0.674	0.675	0.786	0.787	0.792
Banks	70	70	70	70	70	70	70	70
Period	1910-1918							
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Region-Year FE	Y	Y	Y	Y	Y	Y	Y	Y

Note: This table presents annual bank balance sheet items for 1910-1918, adding control variables step by step. Columns (1)-(4) show total assets and columns (5)-(8) deposits, each over 1913 (pre-war) total assets. Within each block, the first column has no controls, the second adds the size of the branch network interacted with Post and the log of the yearly population of the bank's pre-war office locations, the third adds the population-weighted 1909 unemployment rate interacted with Post, and the fourth adds the population-weighted number of banks in the bank's branch locations interacted with Post. The second column is the main specification. Six banks in the balance sheet sample do not report deposits in more than one year and therefore drop out of the deposits columns. Total deposits is the total deposits and current account outstanding balances. Post is a dummy that equals 1 if the year is 1914 or later. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All columns include bank, year and region x year fixed effects, where the regions are north, middle, south and the Randstad (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

Table C.12: Central bank exposure and bank lending: adding control variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Total lending				Illiquid lending			
CB exposure * Post	-0.426***	-0.403**	-0.481***	-0.369**	-0.270***	-0.222**	-0.287***	-0.254**
	(0.157)	(0.163)	(0.158)	(0.185)	(0.102)	(0.090)	(0.087)	(0.114)
# branches * Post		-0.003	-0.007	-0.004		0.011	0.007	0.008
		(0.017)	(0.015)	(0.016)		(0.013)	(0.012)	(0.012)
Log(population), yearly		3.964***	4.033***	4.229***		3.605***	3.663***	3.723***
		(0.976)	(0.879)	(0.802)		(0.674)	(0.606)	(0.586)
Unemployment rate * Post			-0.034	-0.051			-0.028	-0.033
			(0.040)	(0.039)			(0.025)	(0.026)
# Banks * Post				-0.005				-0.001
				(0.003)				(0.002)
Pre-shock mean	0.683	0.683	0.683	0.683	0.474	0.474	0.474	0.474
Observations	523	523	523	523	523	523	523	523
R-squared	0.656	0.682	0.683	0.685	0.714	0.745	0.747	0.747
Banks	70	70	70	70	70	70	70	70
Period	1910-1918							
Bank FE	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Region-Year FE	Y	Y	Y	Y	Y	Y	Y	Y

Note: This table presents annual bank balance sheet items for 1910-1918, adding control variables step by step. Columns (1)-(4) show total lending and columns (5)-(8) illiquid lending, each over 1913 (pre-war) total assets. Within each block, the first column has no controls, the second adds the size of the branch network interacted with Post and the log of the yearly population of the bank's pre-war office locations, the third adds the population-weighted 1909 unemployment rate interacted with Post, and the fourth adds the population-weighted number of banks in the bank's branch locations interacted with Post. The second column is the main specification. Total lending is the sum of all credit lines, repo lending, short term collateralized loans (bills of exchange) and long-term loans; illiquid lending contains credit lines and long-term loans. Post is a dummy that equals 1 if the year is 1914 or later. CB exposure is a continuous treatment variable capturing the fraction of a bank's branches located in the same city as a CB office. All columns include bank, year and region x year fixed effects, where the regions are north, middle, south and the Randstad (North- and South-Holland). Clustered standard errors on the bank level are in parentheses. ***, ** and * indicate significance at 1, 5 and 10% level, respectively.

D Derivation of the Deposit Rate Rule

This appendix derives the deposit rate rule in equation (5). I drop the subscripts j, t wherever possible.

Problem. Each period, bank j chooses its deposit rate $r_{j,t}$ and its non-deposit funding $W_{j,t}$ to maximise profit (3),

$$\Pi = \big(r_t + l_0 - \tfrac{l_1}{2}A\big)A - \big(r_t + \tfrac{h}{2}W\big)W - (r_j + c)\,D, \qquad A = D + W,$$

subject to the deposit demand curve (4), $D = \bar{D}\,[1 + b(r_j - v)]$, where $v = v_{j,t}$ is the common part of the outside option.

Step 1: the marginal value of a deposit guilder. Deposits are a liability: each guilder costs $r_{j,t} + c_{j,t}$. Their value to the bank comes from the assets they fund. Holding W fixed, one additional guilder of deposits raises assets by one guilder ($\partial A/\partial D = 1$), and earns the marginal asset return:

$$\frac{\partial}{\partial A}\Big[\big(r_t + l_0 - \tfrac{l_1}{2}A\big)A\Big] = r_t + l_0 - l_1 A,$$

Net of the cost of the deposit:

$$\frac{\partial \Pi}{\partial D} = \underbrace{r_t + l_0 - l_1 A - c}_{m} - r_j,$$

which defines m in equation (6).

Step 2: non-deposit funding. The first-order condition for W is

$$\frac{\partial \Pi}{\partial W} = (r_t + l_0 - l_1 A) - (r_t + hW) = 0 \quad \Longrightarrow \quad W^* = \frac{l_0 - l_1 D}{l_1 + h}.$$

The bank borrows until the marginal asset return equals the marginal cost of non-deposit funding, so at the optimum $m = r_t + hW^* - c$. Because W is chosen optimally, a small change in W has no first-order effect on profit (envelope theorem), so Step 1 holds whether or not W adjusts. At $t = 1$ the same condition applies with r_1 equal to the discount rate.

Step 3: the deposit rate. Raising r_j attracts $\bar{D}b$ additional deposits, each worth $m - r_j$, but raises the cost of all existing deposits D:

$$\frac{d\Pi}{dr_j} = (m - r_j)\,\bar{D}b - D = 0.$$

Using the demand curve, $D/(\bar{D}b) = 1/b + r_j - v$, so

$$m - r_j = \tfrac{1}{b} + r_j - v \quad \Longrightarrow \quad r_j^* = \tfrac{1}{2}(m + v) - \tfrac{1}{2b},$$

which is equation (5) including the markdown $1/(2b)$.

Step 4: closed form. The marginal value m is evaluated at the optimal balance sheet and therefore depends on D. Substituting W^* from Step 2 into $m = r_t + hW^* - c$,

$$m = r_t - c + \frac{h\, l_0}{l_1 + h} - \gamma D, \qquad \gamma \equiv \frac{l_1 h}{l_1 + h} > 0.$$

The parameter γ measures how much a deposit guilder gains in value when the bank loses one. When deposits fall by one guilder, the bank replaces a share $l_1/(l_1 + h)$ with non-deposit funding and its assets shrink by the remaining $h/(l_1 + h)$, which raises the marginal asset return by $l_1 h/(l_1 + h)$. Both frictions are needed: if non-deposit funding were available at a constant cost ($h = 0$) or returns did not fall with size ($l_1 = 0$), $\gamma = 0$ and m would not depend on deposits. Substituting m and the demand curve into the rule of Step 3 and defining $k \equiv \gamma \bar{D} b > 0$,

$$r_j^* = \frac{(1 + k)\, v + r_t - c + \frac{h\, l_0}{l_1 + h} - \gamma \bar{D} - \frac{1}{b}}{2 + k}.$$

The second-order condition holds: $d^2\Pi/dr_j^2 = -2\bar{D}b - \gamma(\bar{D}b)^2 < 0$.

Step 5: comparative statics. Differentiating the closed form and the demand curve:

$$\frac{\partial r_j^*}{\partial v} = \frac{1 + k}{2 + k} \in \big(\tfrac{1}{2}, 1\big), \qquad \frac{\partial D^*}{\partial v} = \bar{D}b\Big(\frac{\partial r_j^*}{\partial v} - 1\Big) = -\frac{\bar{D}b}{2 + k} < 0,$$
$$\frac{\partial r_j^*}{\partial c} = -\frac{1}{2 + k} < 0, \qquad \frac{\partial D^*}{\partial c} = \bar{D}b\,\frac{\partial r_j^*}{\partial c} = -\frac{\bar{D}b}{2 + k} < 0.$$

A higher outside option raises the deposit rate, but by less than one-for-one, so deposits fall: rate and quantity move in opposite directions (a demand shift). A higher cost of deposits lowers both the rate and deposits (a supply shift). Because m rises as deposits fall, pass-through of the outside option exceeds the $\frac{1}{2}$ in equation (5).

For the balance sheet, $\partial W^*/\partial D = -l_1/(l_1 + h) < 0$ and $\partial A/\partial D = h/(l_1 + h) > 0$: when deposits fall, the bank partly replaces them with non-deposit funding and its assets shrink.

Differences between H and L. Banks H and L share r_t, $\bar{D}$, b, l_0, l_1 and h, and hence k. Because the model is linear, the difference in deposit rates is exact:

$$r_{H,t}^* - r_{L,t}^* = \frac{1 + k}{2 + k}\,\Delta_t \;-\; \frac{1}{2 + k}\,\big(c_{H,t} - c_{L,t}\big).$$

In the crisis ($c_{H,1} = c_{L,1}$, $\Delta_1 > 0$) the difference is positive. After the crisis it is positive if only the outside option persists (case 2), negative if only the cost of deposits differs (case 3), and can be zero if both are present. Deposits at H are lower than at L whenever $\Delta_t > 0$ or $c_{H,t} > c_{L,t}$:

$$D^*_{H,t} - D^*_{L,t} = -\frac{\bar{D}b}{2+k}\left[\Delta_t + (c_{H,t} - c_{L,t})\right].$$